

1. Dado el programa

$$\begin{array}{l} \text{sum} : [\text{Num}] \rightarrow \text{Num} \\ \text{sum}[] \doteq 0 \\ \text{sum}(x \triangleright xs) \doteq x + \text{sum}.xs \end{array}$$

- ¿Qué hace esta función? Escriba en lenguaje natural el **problema** que resuelve.
- Escriba una **especificación** de la función con una expresión cuantificada.
- Verifique** que esta especificación vale para toda lista xs .

$$\begin{aligned} \text{a)} & \text{ Suma todos los elementos de la lista} \\ \text{b)} & \langle \sum i : 0 \leq i < \#xs : xs.i \rangle = \text{sum}.xs \\ \text{c)} & \langle \sum i : 0 \leq i < \#xs : xs.i \rangle \stackrel{H}{=} \langle \sum i : 0 \leq i < \#(x \triangleright xs) : (x \triangleright xs).i \rangle \\ xs = [] & \langle \sum i : 0 \leq i < \#[] : [].i \rangle = \{ \text{Def length y logica} \} \\ & \langle \sum i : i=0 \vee 1 \leq i < \#xs + 1 : (x \triangleright xs).i \rangle \\ & = \{ \text{Def length, rango Vacío} \} \\ & \langle \sum i : \text{false} : [].i \rangle = \{ \text{Partición de rango} \} \\ & = \{ \text{Def rango Vacío} \} \\ & \langle \sum i : i=0 : (x \triangleright xs).i \rangle + \langle \sum i : 1 \leq i < \#xs + 1 : (x \triangleright xs).i \rangle \\ & = \{ \text{Rango unitario, Def index} \} \\ & x + \langle \sum i : 1 \leq i < \#xs + 1 : (x \triangleright xs).i \rangle \\ & = \{ \text{Cambio Var } i := i+1, \text{aritmética} \} \\ & x + \langle \sum i : 0 \leq i < \#xs : (x \triangleright xs).(i+1) \rangle \\ & = \{ \text{Def index} \} \\ & x + \langle \sum i : 0 \leq i < \#xs : xs.i \rangle \\ & = \{ \text{HI} \} \\ & \boxed{x + \text{sum}.xs} \end{aligned}$$

- A partir de la especificación que desarrolló en el ejercicio anterior **derive** la definición de la función. ¿Esta derivación es parecida a la demostración en el ejercicio 1c?

- A partir de las siguientes especificaciones, dar el tipo de cada función y **derivar** las funciones correspondientes.

$$\text{a)} \text{ sum_cuad}.xs = \langle \sum i : 0 \leq i < \#xs : xs.i * xs.i \rangle$$

$$\text{b)} \text{ iga.v}.xs = \langle \forall i : 0 \leq i < \#xs : xs.i = v \rangle$$

$$\text{c)} \text{ fac}.n = n!$$

Ayuda: Recordar que $0! = 1$ y $(n+1)! = (n+1) * n!$ con $n \in \text{Nat}$

$$\text{a)} \text{ sum_cuad} : [\text{Int}] \rightarrow \text{Int}$$

$$\text{sum_cuad}.xs = \langle \sum i : 0 \leq i < \#xs : xs.i * xs.i \rangle \stackrel{H}{=}$$

$$xs = []$$

$$xs = (x \triangleright xs)$$

$$\langle \sum i : 0 \leq i < \#[] : [].i * [].i \rangle$$

$$\langle \sum i : 0 \leq i < \#(x \triangleright xs) : (x \triangleright xs).i * (x \triangleright xs).i \rangle$$

$$= \{ \text{Def length, logica} \}$$

$$= \{ \text{Def length, logica} \}$$

$$\langle \sum i : \text{false} : [].i * [].i \rangle$$

$$\langle \sum i : i=0 : (x \triangleright xs).i * (x \triangleright xs).i \rangle + \langle \sum i : 1 \leq i < 1 + \#xs : (x \triangleright xs).i * (x \triangleright xs).i \rangle$$

$$= \{ \text{Rango Vacío} \}$$

$$= \{ \text{Rango unitario, Def index} \}$$

$$\boxed{0}$$

$$x * x + \langle \sum i : 1 \leq i < 1 + \#xs : (x \triangleright xs).i * (x \triangleright xs).i \rangle$$

Sum_wad: [Int] → Int

Sum_wad.[] = 0

Sum_wad.(x::xs) = x + Sum_wad.xs

= {Cambio variable, aritmética}

$$x + \langle \sum_{i: 0 \leq i < \#xs: (xs[i]) \cdot (xs[i+1])} \rangle$$

= {Def index, HI}

$$x + \text{Sum_wad.xs}$$

$$b) \text{iga.v.xs} = \langle \forall i: 0 \leq i < \#xs: xs[i] = v \rangle \quad (xs): \langle \forall i: 0 \leq i < \#(xs): (xs[i]) = v \rangle$$

xs = [] = {Def length, logica}

= {Def length, logica}

$$\langle \forall i: \text{false}: [][i] = v \rangle$$

= {Rango vacío}

True

iga: Int → [Int] → Bool

iga.v.[] = true

iga.v.(x::xs) = (x=v) ∧ iga.v.xs

$$\langle \forall i: i=0: (xs[i]) = v \rangle \wedge \langle \forall i: 2 \leq i < L + \#xs: (xs[i]) = v \rangle$$

= {Def index, Rango unitario}

$$(x=v) \wedge \langle \forall i: 2 \leq i < L + \#xs: (xs[i]) = v \rangle$$

= {Cambio variable i:=i+1, aritmética}

$$(x=v) \wedge \langle \forall i: 0 \leq i < \#xs: (xs[i+1]) = v \rangle$$

= {Def index, HI}

$$(x=v) \wedge \text{iga.v.xs}$$

$$c) \text{fac.n} = n! \quad \text{fac.(n+1)} = (n+1)! = (n+1) \cdot n!$$

$$n=0 \quad \text{fac.0} = 0! = 1$$

= {HI}

$$\text{fac.0} = 1$$

$$(n+1) \cdot \text{fac.n}$$

fac: Int → Int

fac.0 = 1

fac.n+1 = (n+1) · fac.n

4. Para todos los ítems del ejercicio anterior, dar un ejemplo de uso de la función, es decir: elegir valores concretos para los parámetros y calcular el resultado usando la solución algorítmica obtenida. Las listas deben tener por lo menos tres elementos.

$$c) \quad n = 3$$

$$\text{fac.3} = 2 \cdot \text{fac.3} = 6$$

$$c) \quad xs = [2, 3, 4] \quad \text{sum_wad}([2, 3, 4]) = \langle \sum_{i: 0 \leq i < \#[2, 3, 4]}: [2, 3, 4][i] \cdot [2, 3, 4][i] \rangle$$

= {Def length, rango}

$$\langle \sum_{i: i \in \{0, 1, 2\}}: [2, 3, 4][i] \cdot [2, 3, 4][i] \rangle$$

= {Aplico rango al termino}

$$(2 \cdot 2) + (3 \cdot 3) + (4 \cdot 4) = 4 + 9 + 16 = 29$$

$$b) \quad v = 3, xs = [3, 7, 1]$$

$$\text{iga.3}([3, 7, 1]) = \langle \forall i: 0 \leq i < \#[3, 7, 1]: [3, 7, 1][i] = 3 \rangle$$

= {Def length}

$$\langle \forall i: i \in \{0, 1, 2\}: [3, 7, 1][i] = 3 \rangle$$

= {Aplico el rango al termino}

$$(3=3) \wedge (7=3) \wedge (1=3) = \text{true} \wedge \text{false} \wedge \text{false} = \text{false}$$

5. Expresar en lenguaje natural cada una de las siguientes sentencias formales, donde xs y ys son listas de enteros:

$$a) \langle \exists i: 0 \leq i < \#xs: xs[i] = 0 \rangle$$

$$b) \langle \forall i: 0 \leq i < \#xs: xs[i] \geq 0 \rangle$$

$$c) \langle \exists i: 0 < i < \#xs: xs[i-1] > xs[i] \rangle$$

$$d) \langle \forall x, z: x \in \text{Num} \wedge z \in \text{Num} \wedge x < z: (\exists y: y \in \text{Num}: x < y < z) \rangle$$

$$e) \langle \exists x: x \in \text{Num}: (\forall y: y \in \text{Num}: x \leq y) \rangle$$

$$f) \langle \forall y: y \in \text{Num}: (\exists x: x \in \text{Num}: x \leq y) \rangle \quad \text{¿Cuál es la diferencia con la anterior?}$$

$$g) \#xs \neq \#ys \vee \langle \exists i: 0 \leq i < \#xs \wedge 0 \leq i < \#ys: xs[i] \neq ys[i] \rangle$$

- a) Existe un elemento en la lista que es igual a cero
- b) Para todo elemento de la lista es mayor a cero.
- c) Existe un elemento en la lista tq el anterior es mayor que el siguiente
- d) Para todo x y z que son números, y además x es menor que z . Existe y : (otro número) que se encuentra entre x y z .
- e) Existe x (un número) tq Para todo y (número), x siempre es menor o igual que y .
- f) es igual $\nearrow$??
- g) el length de x s e y s son distintos o, existe un elemento en x s e y s tq son distintos.

6. Decir que expresan las siguientes especificaciones en lenguaje natural y despues derivar las funciones correspondientes.

- ✓ a) $\text{exp}.x.n = x^n$ (con $\text{exp} : \text{Num} \rightarrow \text{Nat} \rightarrow \text{Num}$)
Ayuda: Recordar que $x^0 = 1$ y $x^{n+1} = x * x^n$ con $n \in \text{Nat}$
- ✓ b) $\text{sum_par}.n = \langle \sum i : 0 \leq i \leq n \wedge \text{par}.i : i \rangle$
- ✓ c) $\text{cuántos}.v.xs = \langle \text{N } i : 0 \leq i < \#xs : xs.i = v \rangle$
- d) $\text{busca}.v.xs = \langle \text{Min } i : 0 \leq i < \#xs \wedge xs.i = v : i \rangle$
- e) $\text{mayor_max}.v.xs = \langle \text{Max } i : 0 \leq i < \#xs : xs.i \rangle < v$
 La función derivada no puede usar infinito.
Ayuda: Ver definición 4.11 (pag. 68) del libro.

- a) Calcula el exponente de un número dado.
- b) Suma todos los elementos Pares que van del cero al numero brindado
- c) Cuenta todos los elementos de una lista que son iguales a un numero dado.
- e) busca el máximo elemento de una lista y lo compara con un valor dado, para saber si es menor.

- d) busca los elementos de una lista que son que son iguales a un valor, y compara para obtener el menor.

a) $\text{exp}.x.n = x^n$ (H)

$\Rightarrow n=0 \quad \text{exp}.x.0 = x^0 = 1$

$$\begin{aligned} \text{exp} : \text{Int} \rightarrow \text{Nat} \rightarrow \text{Int} \\ \text{exp}. - 0 &= 1 \\ \text{exp}.x.(n+1) &= x \cdot \text{exp}.x.n \end{aligned}$$

$\text{exp}.x.(n+1) = x^{(n+1)}$

$= x^n \cdot x$

$\stackrel{(H)}{=} \underline{\text{exp}.x.n \cdot x}$

b) $\text{Suma_Par}.n = \langle \sum i : 0 \leq i \leq n \wedge \text{Par}.i : i \rangle$

(C) $\text{Suma_Par}.0 = \langle \sum i : 0 \leq i \leq 0 \wedge \text{Par}.i : i \rangle$

$= \{ \text{lógica} \}$

$\langle \sum i : i=0 \wedge \text{Par}.i : i \rangle$

$= \{ \text{leibniz2} \}$

$\langle \sum i : i=0 \wedge \text{Par}.i : i \rangle$

$= \{ \text{Par}.i := 0, \text{neutro conjunción} \}$

$\langle \sum i : i=0 : i \rangle$

$= \{ \text{rango unitario} \}$

(R) $\text{Suma_Par}.(n+1) = \langle \sum i : 0 \leq i \leq (n+1) \wedge \text{Par}.i : i \rangle$

$= \{ \text{lógica} \}$

$\langle \sum i : (0 \leq i \leq n \vee i=n+1) \wedge \text{Par}.i : i \rangle$

$= \{ \text{Distributividad del } \wedge \text{ con el } \vee, \text{Partición de rango} \}$

$\langle \sum i : 0 \leq i \leq n \wedge \text{Par}.i : i \rangle + \langle \sum i : i=n+1 \wedge \text{Par}.i : i \rangle$

$= \{ H \}$

$\text{Suma_Par}.n + \langle \sum i : i=n+1 \wedge \text{Par}.i : i \rangle$

$= \{ \text{unitario, termino} \}$

$\text{Par}.(n+1) \mid n+1 + \text{Suma_Par}.n$

$$\text{Sum_Par: Int} \rightarrow \text{Int}$$

$$| \neg \text{Par.}(n+1) | \text{Sum_Par.}n$$

$$\text{Sum_Par.}0 = 0$$

$$\text{Sum_Par.}(n+1) \mid (n+1) \bmod 2 = 0 = n+1 + \text{Sum_Par.}n$$

$$\mid (n+1) \bmod 2 \neq 0 = \text{Sum_Par.}n$$

$$c) \text{ Cuantos.}v.xs = \langle \sum i: 0 \leq i < \#xs \wedge xs.i = v: 1 \rangle \quad (A)$$

$$CB \text{ Cuantos.}v.C = \langle \sum i: 0 \leq i < \#C \wedge C.i = v: 1 \rangle$$

$$= \{ \text{lógica, absorbente del } 1 \}$$

$$\langle \sum i: \text{false}: 1 \rangle$$

$$= \{ \text{Rango vacío} \}$$

$$\boxed{10}$$

$$CR \text{ Cuantos.}v.(x \triangleright xs) = \langle \sum i: 0 \leq i < \#(x \triangleright xs) \wedge (x \triangleright xs).i = v: 1 \rangle$$

$$= \{ \text{lógica, Distributividad } \wedge \text{ con } v, \text{ Partición de rango} \}$$

$$\langle \sum i: i=0 \wedge (x \triangleright xs).i = v: 1 \rangle + \langle \sum i: 2 \leq i < \#(x \triangleright xs) \wedge (x \triangleright xs).i = v: 1 \rangle$$

$$= \{ \text{Def length, aritmética, Def index} \}$$

$$\langle \sum i: i=0 \wedge (x \triangleright xs).i = v: 1 \rangle + \langle \sum i: 0 \leq i < \#xs \wedge xs.i = v: 1 \rangle$$

$$= \{ HI \}$$

$$\langle \sum i: i=0 \wedge (x \triangleright xs).i = v: 1 \rangle + \text{Cuantos.}v.xs$$

$$= \{ \text{Rango unitario Conteo} \}$$

$$\left(\begin{array}{l} x=v \rightarrow 1 + \text{Cuantos.}v.xs \\ \mid x \neq v \rightarrow \text{Cuantos.}v.xs \end{array} \right)$$

$$\text{Cuantos.}: \text{Int} \rightarrow [\text{Int}] \rightarrow \text{Int}$$

$$\text{Cuantos.}v.C = 0$$

$$\text{Cuantos.}v.(x \triangleright xs) \left(\begin{array}{l} (x=v) \rightarrow 1 + \text{Cuantos.}v.xs \\ \mid (x \neq v) \rightarrow \text{Cuantos.}v.xs \end{array} \right)$$

$$\mid (x \neq v) \rightarrow \text{Cuantos.}v.xs$$

$$d) \text{ busca.}v.xs = \langle \text{Min.}i: 0 \leq i < \#xs \wedge xs.i = v: i \rangle \quad (H)$$

$$CB \text{ busca.}v.C = \langle \text{Min.}i: 0 \leq i < \#C \wedge C.i = v: i \rangle$$

$$= \{ \text{Def length, lógica, absorbente} \}$$

$$\langle \text{Min.}i: \text{false}: i \rangle$$

$$= \{ \text{Rango vacío} \}$$

$$\boxed{1+10}$$

$$CR \text{ busca.}v.(x \triangleright xs) = \langle \text{Min.}i: 0 \leq i < \#(x \triangleright xs) \wedge (x \triangleright xs).i = v: i \rangle$$

$$= \{ \text{lógica, distributividad } \wedge \text{ con } v, \text{ Partición de rango} \}$$

$$\langle \text{Min.}i: i=0 \wedge (x \triangleright xs).i = v: i \rangle \text{Min} \langle \text{Min.}i: 2 \leq i < \#(x \triangleright xs) \wedge (x \triangleright xs).i = v: i \rangle$$

$$= \{ \text{Cambio variable } i: i+1, \text{ index} \}$$

$$\langle \text{Min.}i: i=0 \wedge (x \triangleright xs).i = v: i \rangle \text{Min} \langle \text{Min.}i: 0 \leq i < \#xs \wedge xs.i = v: i+1 \rangle$$

$$= \{ \text{termino constante, HI} \}$$

$$\langle \text{Min.}i: i=0 \wedge (x \triangleright xs).i = v: i \rangle \text{Min} (\text{busca.}v.xs + 1)$$

$$= \{ \text{Rango unit} \}$$

$(x == v = 0$
 $\exists x \neq v = \text{busca.v.xs} + 1$
 $)$

~~$\text{busca: Int} \rightarrow [\text{Int}] \rightarrow \text{Int}$
 $\text{busca.v.c} = \text{maxBound}$
 $\text{busca.v.(x \> xs)} \begin{cases} x = v \rightarrow 0 \\ \exists x \neq v \rightarrow \text{busca.v.xs} + 1 \end{cases}$~~

$\text{e) mayor_max.v.xs} = \langle \text{Max.i: } 0 \leq i < \#xs : xs.i \rangle$

$\text{c) mayor_max.v.c} = \langle \text{Max.i: } 0 \leq i < \#c : c.i \rangle$

$= \{ \text{Def length, logica, Rango Vacio} \}$

$[-\infty] \text{ (Min Bound)}$

$\text{mayor_max: Int} \rightarrow [\text{Int}] \rightarrow \text{Int}$

$\text{mayor_max.n.c} = \text{maxBound}$

$\text{mayor_max.n.(x \> xs)} = x \text{ max mayor_max.v.xs}$

(H)

$\text{c) mayor_max.v.(x \> xs)} = \langle \text{Max.i: } 0 \leq i < \#(x \> xs) : (x \> xs).i \rangle$

$= \{ \text{Def length, logica} \}$

$\langle \text{Max.i: } i=0 : (x \> xs).i \rangle \text{ max } \langle \text{Max.i: } 1 \leq i < \#(x \> xs) : (x \> xs).i \rangle$

$= \{ \text{Rango unitario, Def index} \}$

$x \text{ max } \langle \text{Max.i: } 1 \leq i < \#(x \> xs) : (x \> xs).i \rangle$

$= \{ \text{Def de length, Cambio de Variable } i := i+1, \text{ aritmética} \}$

$x \text{ max } \langle \text{Max.i: } 0 \leq i < \#xs : (x \> xs).(i+1) \rangle$

$= \{ \text{Def index} \}$

$x \text{ max } \langle \text{Max.i: } 0 \leq i < \#xs : xs.i \rangle$

$= \{ \text{HI} \}$

$x \text{ max mayor_max.v.xs} < n$

$??$

8. Derivar las siguientes funciones.

a) $f: \text{Num} \rightarrow \text{Nat} \rightarrow \text{Num}$ computa la suma de potencias de un número, esto es

$$f.x.n = \langle \sum i : 0 \leq i < n : x^i \rangle$$

b) $\cos: \text{Nat} \rightarrow \text{Num} \rightarrow \text{Num}$ computa la aproximación de la función trigonométrica coseno:

$$\cos.n.x = \langle \sum i : 0 \leq i \leq n : (-x)^{2*i} / (2*i)! \rangle$$

c) $f: \text{Nat} \rightarrow \text{Nat}$ computa el cubo de un número natural x utilizando únicamente sumas. La especificación es muy simple: $f.x = x^3$.

Ayuda: Usar inducción y modularización varias veces

a)

$$f.x.n = \langle \sum i : 0 \leq i \leq n : x^i \rangle \quad (H)$$

CD) $f.x.0 = \langle \sum i : 0 \leq i \leq 0 : x^i \rangle$

= {lógica, Rango vacío}

[1]

$$f: \text{Int} \rightarrow \text{Nat} \rightarrow \text{Int}$$

$$f.x.0 = 0$$

$$f.x.(n+1) = \exp.x.n + f.x.n$$

CP) $f.x.(n+1) = \langle \sum i : 0 \leq i < (n+1) : x^i \rangle$

= {lógica}

$$\langle \sum i : i=n \vee 0 \leq i < n : x^i \rangle$$

= {Partición de rango}

$$\langle \sum i : i=n : x^i \rangle + \langle \sum i : 0 \leq i < n : x^i \rangle$$

= {Rango unitario} (H)

$$x^n + f.x.n$$

= {Modularizo: ej. 6a)}

$$\boxed{\exp.x.n + f.x.n}$$

b) $\cos: \text{Nat} \rightarrow \text{Num} \rightarrow \text{Num}$

$$\cos.n.x = \langle \sum i : 0 \leq i < n : (-1)^{2*i} / (2*i)! \rangle \quad (H)$$

CD) $n=0$

$$\cos.0.x = \langle \sum i : 0 \leq i < 0 : (-1)^{2*i} / (2*i)! \rangle$$

= {Rango vacío.}

$$\langle \sum i : \text{false} : (-1)^{2*i} / (2*i)! \rangle$$

= {Definición Rango vacío de la sumatoria}

[10]

CP) $\cos.(n+1).x = \langle \sum i : 0 \leq i < n+1 : (-1)^{2*i} / (2*i)! \rangle$

= {lógica}

$$\langle \sum i : 0 \leq i < n \vee i=n+1 : (-1)^{2*i} / (2*i)! \rangle$$

= {Partición de rango}

$$\langle \sum i : 0 \leq i < n : (-1)^{2*i} / (2*i)! \rangle + \langle \sum i : i=n+1 : (-1)^{2*i} / (2*i)! \rangle$$

= {HI} $\cos.n.x + \langle \sum i : i=n+1 : (-1)^{2*i} / (2*i)! \rangle$

= {Rango unitario}

$$\cos.n.x + \frac{(-x)^{2(i+1)}}{(2 \cdot (i+1))!}$$

$$= \{ \text{Modularización Función exponencial } i=n \}$$

$$\cos.n.x + \exp.(-x) \cdot \frac{(2i+2)}{(2i+1)!}$$

$$= \{ \text{Modularización Función factorial } i=n \}$$

$$\cos.n.x + \exp.(-x) \cdot \frac{(2n+2)}{\text{fac.}(2n+2)}$$

$$\cos : \text{Nat} \rightarrow \text{Num} \rightarrow \text{Nat}$$

$$\cos.0.x \doteq 1$$

$$\cos.(n+1).x = \cos.n.x + \exp.(-x) \cdot \frac{(2n+2)}{\text{fac.}(2n+2)}$$

$$\exp : \text{Int} \rightarrow \text{Nat} \rightarrow \text{Int}$$

$$\exp.0 = 1$$

$$\exp.x.(n+1) \doteq x \cdot \exp.x.n$$

$$\text{fac} : \text{Int} \rightarrow \text{Int}$$

$$\text{fac}.0 \doteq 1$$

$$\text{fac.}(n+1) \doteq (n+1) \cdot \text{fac}.n$$

$$(n+1)! \doteq (n+1) \cdot n!$$

$$f.n$$

$$c) f : \text{Nat} \rightarrow \text{Nat}$$

$$f.x \doteq x^3$$

$$\textcircled{CB} \quad \textcircled{CH} f.(x+1) \doteq (x+1)^3$$

$$f.0 = 0^3$$

$$= \{ \text{aritmética} \}$$

$$\textcircled{ID},$$

$$= \{ \text{Aritmética} \}$$

$$(x^2 + 2x + 1)(x+1)$$

$$= \{ \text{distributividad} \}$$

$$x^3 + 3x^2 + 3x + 1$$

$$= \{ \text{HI} \} \{ \text{Modulariz } \text{Cuadrado } x \doteq x^2 \}$$

$$f.x + \text{Cuadrado}.x + \text{Cuadrado}.x + \text{Cuadrado}.x + x + x + x + 1$$

$$f : \text{Nat} \rightarrow \text{Nat}$$

$$f.0 \doteq 0$$

$$f.(x+1) \doteq f.x + \text{Cuadrado}.x + \text{Cuadrado}.x + \text{Cuadrado}.x + x + x + x + 1$$

$$\text{Cuadrado } x \doteq x^2 \quad \textcircled{H}$$

$$\bullet \text{Cuadrado} : \text{Int} \rightarrow \text{Int}$$

$$\text{Cuadrado}.0 \doteq 0$$

$$\text{Cuadrado}(x+1) \doteq \text{Cuadrado}.x + x + x + 1$$

$$x^2 + x + x + 1$$

$$\textcircled{H}$$

$$\text{Cuadrado} : \text{Int} \rightarrow \text{Int}$$

$$\text{Cuadrado}.0 \doteq 0$$

$$\text{Cuadrado}(x+1) \doteq \text{Cuadrado}.x + x + x + 1$$

9. Especificar y derivar un programa que determine si un elemento ocurre un número par de veces en una lista.

$$\text{Contar_Cuantas}.xs.v = \langle \sum i : 0 \leq i < \#xs \wedge xs.i = v : 1 \rangle \bmod 2 = 0$$

$$= \{ \text{Modularizo} \}$$

$$\text{Contar_Cuantas}.xs.v = \langle \sum i : 0 \leq i < \#xs \wedge xs.i = v : 1 \rangle \quad \textcircled{H}$$

$$\textcircled{CB} \text{Contar_Cuantas}[].v = \langle \sum i : 0 \leq i < \#[].i = v : 1 \rangle$$

$$= \{ \text{lógica} \}$$

$$\langle \sum i : \text{false} \wedge [].i = v : 1 \rangle$$

$$\begin{aligned}
& \text{Contar_cuantas} \cdot (xs) \cdot v \equiv \langle \sum i: 0 \leq i < \#(xs) \wedge (xs)[i] = v : 1 \rangle \\
& = \{ \text{lógica, Prop. distributiva} \} \\
& \langle \sum i: i=0 \wedge (xs)[i] = v : 2 \rangle + \langle \sum i: 1 \leq i < \#(xs) \wedge (xs)[i] = v : 1 \rangle \\
& = \{ \text{Def. indexar} \} \\
& \quad + \langle \sum i: 1 \leq i < \#(xs) \wedge (xs)[i] = v : 1 \rangle \\
& = \{ \text{Def length, Cambio de variable } i := i+1, \text{aritmética} \} \\
& \quad + \langle \sum i: 0 \leq i < \#xs \wedge (xs)[i+1] = v : 1 \rangle \\
& = \{ \text{caso recursivo indexar} \} \\
& \quad + \langle \sum i: 0 \leq i < \#xs \wedge xs[i] = v : 1 \rangle \\
& = \{ \text{Leibniz Def, Rango unitario} \} \\
& \quad \left(\begin{aligned} & x = v \rightarrow 1 + \text{Contar_cuantas} \cdot xs \cdot v \\ & \vee x \neq v \rightarrow \text{Contar_cuantas} \cdot xs \cdot v \end{aligned} \right)
\end{aligned}$$

Contar_cuantas : [a] → a → Bool

Contar_cuantas.[] . v ≡ true

$$\text{Contar_cuantas} \cdot (xs) \cdot v \equiv \left(\begin{aligned} & x = v \rightarrow (1 + \text{Contar_cuantas} \cdot xs \cdot v) \bmod 2 = 0 \\ & \vee x \neq v \rightarrow \text{Contar_cuantas} \cdot xs \cdot v \end{aligned} \right)$$

10. Especificar y derivar la función iguales : [A] → Bool, que determina si los elementos de una lista de tipo A son todos iguales entre si.

Ayuda: para la especificación use un ejercicio del práctico anterior.

Ayuda: para la derivación use el esquema inductivo del ejemplo 10.1 del libro.

$$\begin{aligned}
& \text{iguales} \cdot xs \equiv \langle \forall i, j: 0 \leq i < j < \#xs : xs[i] = xs[j] \rangle \\
& \text{iguales} [] \equiv \langle \forall i, j: 0 \leq i < j < \#[] : [][i] = [][j] \rangle \\
& = \{ \text{Rango vacío} \} \\
& \langle \forall i, j: \text{False} : [][i] = [][j] \rangle \quad \text{CP} \text{ iguales}(xs \vee xs) \equiv \langle \forall i, j: 0 \leq i < j < \#(xs \vee xs) : (xs \vee xs)[i] = (xs \vee xs)[j] \rangle \\
& = \{ \text{Def rango vacío} \} \quad = \{ \text{lógica, partición de rango} \} \\
& \quad \text{True} \\
& \langle \forall i, j: (i=0 \wedge j=1) : (xs \vee xs)[i] = (xs \vee xs)[j] \rangle \wedge \langle \forall i, j: 1 \leq i < j < \#(xs \vee xs) : (xs \vee xs)[i] = (xs \vee xs)[j] \rangle \\
& = \{ \text{Regla de Leibniz, Rango unitario} \} \\
& \quad \langle (xs \vee xs)[0] = (xs \vee xs)[1] \wedge \forall i, j: 1 \leq i < j < \#(xs \vee xs) : (xs \vee xs)[i] = (xs \vee xs)[j] \rangle \\
& = \{ \text{Def index} \} \\
& \quad (x = y) \wedge \langle \forall i, j: 1 \leq i < j < \#(xs \vee xs) : (xs \vee xs)[i] = (xs \vee xs)[j] \rangle \\
& = \{ \text{Def length, Cambio de variable } i := i+1, j := j+1, \text{aritmética} \}
\end{aligned}$$

$$(x=y) \wedge \langle \forall i,j: 0 \leq i < j < \#(x \triangleright x) : (x \triangleright x)_i = (x \triangleright x)_j \rangle$$

$$= \{ \text{Def indexar} \}$$

$$(x=y) \wedge \langle \forall i,j: 0 \leq i < j < \#(y \triangleright x) : (y \triangleright x)_i = (y \triangleright x)_j \rangle$$

$$= \{ \text{HI} \} \quad \boxed{(x=y) \wedge \text{iguales.}(y \triangleright x)}$$

$$\textcircled{CB} \text{ iguales. } x \equiv \langle \forall i,j: 0 \leq i < j < \#x : x_i = x_j \rangle$$

$$= \{ \text{Def length} \}$$

$$\langle \forall i,j: 0 \leq i < j < 1 : x_i = x_j \rangle$$

$$= \{ \text{Por logica, Rango vacio} \} \quad \boxed{\text{True}}$$

$$\text{Iguales. } [a] \rightarrow \text{Bool}$$

$$\text{Iguales. } [] \equiv \text{true}$$

$$\text{Iguales. } [x] \equiv \text{true}$$

$$\text{Iguales. } (x \triangleright y \triangleright x) \equiv (x=y) \wedge \text{iguales.}(y \triangleright x)$$

11. Especificar formalmente utilizando cuantificadores cada una de las siguientes funciones descriptas informalmente. Luego, derivar soluciones algorítmicas para cada una.

a) *minimo* : $[Int] \rightarrow Int$, que calcula el mínimo elemento de una lista **no vacía** de enteros.

Nota: La función no debe devolver $\pm\infty$.

b) *creciente* : $[Int] \rightarrow \text{Bool}$, que determina si los elementos de una lista de enteros están ordenados en forma creciente.

c) *prod* : $[Num] \rightarrow [Num] \rightarrow Num$, que calcula el producto entre pares de elementos en iguales posiciones de las listas y suma estos resultados (producto punto). Si las listas tienen distinto tamaño se opera hasta la última posición de las más chica.

$$a) \text{ minimo. } xs \equiv \langle \text{Min. } i: 0 \leq i < \#xs \wedge xs \neq [] : xs_i \rangle \textcircled{H}$$

$$\textcircled{CB} \text{ minimo } [x] \equiv \langle \text{Min. } i: 0 \leq i < \#x \wedge x \neq [] : x_i \rangle$$

$$= \{ \text{Def length, logica} \}$$

$$\langle \text{Min. } i: i=0 \wedge \text{true} : [x]_i \rangle$$

$$= \{ \text{Rango unitario} \}$$

$$= \{ \text{Def indexar} \} \quad \boxed{[x]_0}$$

$$\textcircled{CB} \text{ minimo. } (x \triangleright y \triangleright x) \equiv \langle \text{Min. } i: 0 \leq i < \#(x \triangleright y \triangleright x) \wedge (x \triangleright y \triangleright x) \neq [] : (x \triangleright y \triangleright x)_i \rangle$$

$$= \{ \text{logica, Distributividad} \}$$

$$\langle \text{Min. } i: i=0 \wedge (x \triangleright y \triangleright x) \neq [] : (x \triangleright y \triangleright x)_i \rangle \min \langle \text{Min. } i: 1 \leq i < \#(x \triangleright y \triangleright x) \wedge (x \triangleright y \triangleright x) \neq [] : (x \triangleright y \triangleright x)_i \rangle$$

$$= \{ \text{Rango unitario} \}$$

$$(x \triangleright y \triangleright x)_0 \min \langle \text{Min. } i: 1 \leq i < \#(x \triangleright y \triangleright x) \wedge (x \triangleright y \triangleright x) \neq [] : (x \triangleright y \triangleright x)_i \rangle$$

$$= \{ \text{Def index} \}$$

$$x \min \langle \text{Min. } i: 1 \leq i < \#(x \triangleright y \triangleright x) \wedge (x \triangleright y \triangleright x) \neq [] : (x \triangleright y \triangleright x)_i \rangle$$

$$= \{ \text{Idempotencia, def length, Cambio variable } i := i+1, \text{ aritmetica} \}$$

$$x \min < \text{Min}.i: 0 \leq i < \#(y \triangleright x_s) : (x \triangleright y \triangleright x_s).(i+1) >$$

= {Def index}

$$x \min < \text{Min}.i: 0 \leq i < \#(y \triangleright x_s) : (y \triangleright x_s).i >$$

= {HI}

$$x \min \minimo.(y \triangleright x_s)$$

$$\text{minimo} : [\text{Int}] \rightarrow \text{Int}$$

$$\text{minimo}[x] \doteq x$$

$$\text{minimo}(x \triangleright y \triangleright x_s) \doteq x \min \minimo(y \triangleright x_s)$$

$$b) \text{creciente}.xs \doteq \langle \forall i,j: 0 \leq i < j < \#xs : xs.i \leq xs.j \rangle \quad (H)$$

$$\text{creciente}.[] \doteq \langle \forall i,j: 0 \leq i < j < \#[] : [] . i \leq [] . j \rangle$$

= {Def length, Rango vacio}

$$\langle \forall i,j: \text{false} : \#[] . i \leq [] . j \rangle$$

= {Def nuevo Paratodo(V)}

True

$$\text{creciente}.[] \doteq \langle \forall i,j: 0 \leq i < j < \#[] : [] . i \leq [] . j \rangle$$

= {Def length, Rango vacio, logica}

$$\langle \forall i,j: \text{false} : \#[] . i \leq [] . j \rangle$$

= {Def nuevo Paratodo(V)}

True

$$\text{creciente}(x \triangleright y \triangleright x_s) \doteq \langle \text{Min}.i,j: 0 \leq i < j < \#(x \triangleright y \triangleright x_s) : (x \triangleright y \triangleright x_s).i \leq (x \triangleright y \triangleright x_s).j \rangle$$

= {logica}

$$\langle \text{Min}.i,j: (i=0 \wedge j=1) \vee 1 \leq i < j < \#(x \triangleright y \triangleright x_s) : (x \triangleright y \triangleright x_s).i \leq (x \triangleright y \triangleright x_s).j \rangle$$

= {Partición de rango}

$$\langle \text{Min}.i,j: i=0 \wedge j=1 : (x \triangleright y \triangleright x_s).i \leq (x \triangleright y \triangleright x_s).j \rangle \wedge \text{Min}.i,j: 1 \leq i < j < \#(x \triangleright y \triangleright x_s) : (x \triangleright y \triangleright x_s).i \leq (x \triangleright y \triangleright x_s).j$$

= {Regla de leibniz 2}

$$(x \triangleright y \triangleright x_s).0 \leq (x \triangleright y \triangleright x_s).1 \wedge \text{Min}.i,j: 1 \leq i < j < \#(x \triangleright y \triangleright x_s) : (x \triangleright y \triangleright x_s).i \leq (x \triangleright y \triangleright x_s).j$$

= {Def length, Cambio Variable i:=i+1, j:=j+1, aritmetica}

$$(x \triangleright y \triangleright x_s).0 \leq (x \triangleright y \triangleright x_s).1 \wedge \text{Min}.i,j: 0 \leq i < j < \#(y \triangleright x_s) : (x \triangleright y \triangleright x_s).(i+1) \leq (x \triangleright y \triangleright x_s).(j+1)$$

= {Def indexar}

$$(x \triangleright y \triangleright x_s).0 \leq (x \triangleright y \triangleright x_s).1 \wedge \text{Min}.i,j: 0 \leq i < j < \#(y \triangleright x_s) : (y \triangleright x_s).i \leq (y \triangleright x_s).j$$

= {HI}

$$(x \triangleright y \triangleright x_s).0 \leq (x \triangleright y \triangleright x_s).1 \wedge \text{creciente}.(y \triangleright x_s)$$

= {Def indexar}

$$(x \leq y) \wedge \text{creciente}.(y \triangleright x_s)$$

$$\text{creciente} : [\text{Int}] \rightarrow \text{Bool}$$

$$\text{creciente}.[] \doteq \text{true}$$

$$\text{creciente}[x] \doteq \text{true}$$

$$\text{creciente}(x \triangleright y \triangleright x_s) \doteq x \leq y \wedge \text{creciente}.(y \triangleright x_s)$$

$$c) \text{Prod}.xs.y_s \equiv \langle \exists i: 0 \leq i < \#xs \wedge i < \#ys : xs.i \cdot ys.i \rangle \quad (H)$$

$$\text{Prod}.[] . ys \equiv \langle \exists i: 0 \leq i < \#[] \wedge i < \#ys : [] . i \cdot ys.i \rangle$$

= {Def length, absorbente del $\wedge$ }

$$\langle \exists i: \text{False} : [] . i \cdot ys.i \rangle$$

= {Rango vacío}

$\square_{//}$

$$\text{Prod}.xs.[] \equiv \langle \exists i: 0 \leq i < \#xs \wedge i < \#[] : xs.i \cdot [] . i \rangle$$

= {i demPotencia False $\wedge$ }

$$\langle \exists i: \text{False} : xs.i \cdot [] . i \rangle$$

= {Rango vacío}

$\square_{//}$

$$\text{Prod}.(x \bowtie xs).(y \bowtie ys) \equiv \langle \exists i: 0 \leq i < \#(x \bowtie xs) \wedge i < \#(y \bowtie ys) : (x \bowtie xs).i \cdot (y \bowtie ys).i \rangle$$

= {Def length, Distributividad, Partición de rango}

$$\langle \exists i: i = 0 \wedge i < \#(y \bowtie ys) : (x \bowtie xs).i \cdot (y \bowtie ys).i \rangle + \langle \exists i: 2 \leq i < \#(x \bowtie xs) \wedge i < \#(y \bowtie ys) : (x \bowtie xs).i \cdot (y \bowtie ys).i \rangle$$

= {IdemPotencia, Rango unitario}

$$(x \bowtie xs).0 \cdot (y \bowtie ys).0 + \langle \exists i: 2 \leq i < \#(x \bowtie xs) \wedge i < \#(y \bowtie ys) : (x \bowtie xs).i \cdot (y \bowtie ys).i \rangle$$

= {Def length, Cambio de Variable $i = i+1$ }

$$(x \bowtie xs).0 \cdot (y \bowtie ys).0 + \langle \exists i: 0 \leq i < \#xs \wedge i < \#ys : (x \bowtie xs).(i+1) \cdot (y \bowtie ys).(i+1) \rangle$$

= {Def Indexar}

$$x \cdot y + \langle \exists i: 0 \leq i < \#xs \wedge i < \#ys : xs.i \cdot ys.i \rangle$$

= {HI}

$$x \cdot y + \text{Prod}.xs.y_s$$

$$\text{Prod} : [\text{Num}] \rightarrow [\text{Num}] \rightarrow \text{Num}$$

$$\text{Prod}.[] . ys \equiv 0$$

$$\text{Prod}.xs.[] \equiv 0$$

$$\text{Prod}.(x \bowtie xs).(y \bowtie ys) \equiv x \cdot y + \text{Prod}.xs.y_s$$

12. A partir de las siguientes especificaciones expresar en lenguaje natural qué devuelven las funciones, agregarles su tipo y derivarlas:

a) $\text{psum}.xs = \langle \forall i: 0 \leq i \leq \#xs : \text{sum}.(xs \uparrow i) \geq 0 \rangle$, con sum la función del ejercicio 1.

b) $\text{sum.ant}.xs = \langle \exists i: 0 \leq i < \#xs : xs.i = \text{sum}.(xs \uparrow i) \rangle$

c) $\text{sum}8.xs = \langle \exists i: 0 \leq i \leq \#xs : \text{sum}.(xs \uparrow i) = 8 \rangle$.

d) $f.xs = \langle \text{Max } i: 0 \leq i < \#xs \wedge \text{sum}.(xs \uparrow i) = \text{sum}.(xs \downarrow i) : i \rangle$.

$$\text{Sum} : [\text{Int}] \rightarrow \text{Int}$$

$$\text{Sum}.[] \equiv 0$$

$$\text{Sum}(x \bowtie xs) \equiv x + \text{Sum}.xs$$

$$a) \text{Psum}.xs \equiv \langle \forall i: 0 \leq i \leq \#xs : \text{sum}.(xs \uparrow i) \geq 0 \rangle \quad (H)$$

$$\text{Psum}.[] \equiv \langle \forall i: 0 \leq i \leq \#[] : \text{sum}.([] \uparrow i) \geq 0 \rangle$$

= {Def length, rango unitario}

$$\text{sum}.([] \uparrow 0) \geq 0$$

= {Def Take}

$$\text{Sum } [] \geq 0$$

= {CB Sum}

= {logica} $\frac{0 \geq 0}{\boxed{\text{true}}}$

$$\textcircled{CB} \text{ Psum. } (x \triangleright xs) = \langle \forall i: 0 \leq i \leq \#(x \triangleright xs) : \text{sum}(x \triangleright xs) \uparrow i \geq 0 \rangle$$

= {logica, Partición de rango}

$$\langle \forall i: i = 0 : \text{sum}(x \triangleright xs) \uparrow i \geq 0 \rangle \wedge \langle \forall i: 1 \leq i \leq \#(x \triangleright xs) : \text{sum}(x \triangleright xs) \uparrow i \geq 0 \rangle$$

= {Cambio de variable $i := i+1$, Def length, aritmética}

$$\textcircled{C} \langle \forall i: 0 \leq i \leq \#xs : \text{sum}(x \triangleright xs) \uparrow (i+1) \geq 0 \rangle$$

= {Def Take}

$$\langle \forall i: 0 \leq i \leq \#xs : \text{sum}(x \triangleright (xs \uparrow i)) \geq 0 \rangle$$

= {Def Sum}

$$\langle \forall i: 0 \leq i \leq \#xs : x + \text{sum}(xs \uparrow i) \geq 0 \rangle$$

= {Debo de generalizar, no puedo llegar a la H}

$$\text{Psum. } xs = \text{gsum. } 0. xs$$

$$\text{gsum. } n. xs = \langle \forall i: 0 \leq i \leq \#xs : n + \text{sum}(xs \uparrow i) \geq 0 \rangle \quad \textcircled{H}$$

$$\textcircled{CB} \text{ gsum. } n. [] = \langle \forall i: 0 \leq i \leq \#[] : n + \text{sum}([] \uparrow i) \geq 0 \rangle$$

= {Rango unitario}

$$n + \text{sum}([] \uparrow i) \geq 0$$

= {Def take, Def sumar}

$$\boxed{n \geq 0}$$

$$\textcircled{CB} \text{ gsum. } n. (x \triangleright xs) = \langle \forall i: 0 \leq i \leq \#(x \triangleright xs) : n + \text{sum}(x \triangleright xs) \uparrow i \geq 0 \rangle$$

= {logica, Partición de rango}

$$\langle \forall i: i = 0 : n + \text{sum}(x \triangleright xs) \uparrow i \geq 0 \rangle \wedge \langle \forall i: 1 \leq i \leq \#(x \triangleright xs) : n + \text{sum}(x \triangleright xs) \uparrow i \geq 0 \rangle$$

= {Rango unitario}

$$n + \text{sum}(x \triangleright xs) \uparrow 0 \geq 0 \wedge \langle \forall i: 1 \leq i \leq \#(x \triangleright xs) : n + \text{sum}(x \triangleright xs) \uparrow i \geq 0 \rangle$$

= {Cambio de variable $i := i+1$, aritmética, Def length}

$$n + \text{sum}(x \triangleright xs) \uparrow 0 \geq 0 \wedge \langle \forall i: 0 \leq i \leq \#xs : n + \text{sum}(x \triangleright xs) \uparrow (i+1) \geq 0 \rangle$$

= {Def take}

$$n + \text{sum. } [] \geq 0 \wedge \langle \forall i: 0 \leq i \leq \#xs : n + \text{sum}(x \triangleright (xs \uparrow i)) \geq 0 \rangle$$

= {Def Sum}

$$n \geq 0 \wedge \langle \forall i: 0 \leq i \leq \#xs : n + x + \text{sum}(xs \uparrow i) \geq 0 \rangle$$

= {Asociatividad}

$$n \geq 0 \wedge \langle \forall i: 0 \leq i \leq \#xs : (n + x) + \text{sum}(xs \uparrow i) \geq 0 \rangle$$

= {HI}

$$n \geq 0 \wedge \text{gsum. } (n+x). xs$$

$$\text{gsum. } \text{Num} \rightarrow [\text{Num}] \rightarrow \text{Bool}$$

$$\text{gsum. } n. [] = n \geq 0$$

$$\text{gsum. } n. (x \triangleright xs) = n \geq 0 \wedge \text{gsum. } (n+x). xs$$

$$b) \text{sum_ant. xs} = \langle \exists i: 0 \leq i < \#xs: xs.i = \text{sum.}(xs \uparrow i) \rangle$$

$$\text{sum_ant. C} = \langle \exists i: 0 \leq i < \#C: C.i = \text{sum.}(C \uparrow i) \rangle$$

= {Def length, logica}

$$\langle \exists i: \text{False}: C.i = \text{sum.}(C \uparrow i) \rangle$$

= {Rango vacio}

False

= {logica, Partición de rango}

$$\langle \exists i: i = 0: (xs).i = \text{sum.}((xs) \uparrow i) \rangle \vee \langle \exists i: 1 \leq i < \#(xs): (xs).i = \text{sum.}((xs) \uparrow i) \rangle$$

= {Def de length, cambio de variable $i := i+1$, aritmetica}

$$\textcircled{D} \langle \exists i: 0 \leq i < \#xs: (xs).i = \text{sum.}((xs) \uparrow (i+1)) \rangle$$

= {Def index, Def take}

$$\langle \exists i: 0 \leq i < \#xs: xs.i = \text{sum.}(xs \uparrow i) \rangle$$

= {Def sum}

$$\langle \exists i: 0 \leq i < \#xs: xs.i = x + \text{sum.}(xs \uparrow i) \rangle$$

= {generaliz0}

$$\text{sum_ant. n. xs} = \langle \exists i: 0 \leq i < \#xs: xs.i = n + \text{sum.}(xs \uparrow i) \rangle$$

$$\textcircled{D} \text{sum_ant. n. C} = \langle \exists i: 0 \leq i < \#C: C.i = n + \text{sum.}(C \uparrow i) \rangle$$

= {logica, Rango vacio}

False

$$\textcircled{R} \text{sum_ant. n. (xs)} = \langle \exists i: 0 \leq i < \#(xs): (xs).i = n + \text{sum.}((xs) \uparrow i) \rangle$$

= {logica, Partición de rango}

$$\langle \exists i: i = 0: (xs).i = n + \text{sum.}((xs) \uparrow i) \rangle \vee \langle \exists i: 1 \leq i < \#(xs): (xs).i = n + \text{sum.}((xs) \uparrow i) \rangle$$

= {Rango unitario}

$$(xs).0 = n + \text{sum.}((xs) \uparrow 0) \vee \langle \exists i: 1 \leq i < \#(xs): (xs).i = n + \text{sum.}((xs) \uparrow i) \rangle$$

= {Def length, cambio de variable $i := i+1$, aritmetica}

$$(xs).0 = n + \text{sum.}((xs) \uparrow 0) \vee \langle \exists i: 0 \leq i < \#xs: (xs).(i+1) = n + \text{sum.}((xs) \uparrow (i+1)) \rangle$$

= {Def take}

$$(xs).0 = n + \text{sum. C} \vee \langle \exists i: 0 \leq i < \#xs: (xs).(i+1) = n + \text{sum.}(xs \uparrow i) \rangle$$

= {Def index, def sum}

$$(x = n) \vee \langle \exists i: 0 \leq i < \#xs: xs.i = n + x + \text{sum.}(xs \uparrow i) \rangle$$

= {Asociatividad, HI}

$$x = n \vee \text{sum_ant. (n+x). xs.}$$

$$\text{sum_ant}: \text{Num} \rightarrow [\text{Num}] \rightarrow \text{Bool}$$

$$\text{sum_ant. n. C} = \text{False}$$

$$\text{sum_ant. n. (xs)} = x = n \vee \text{sum_ant. (n+x). xs}$$

$$c) \text{sum8}.xs = \langle \exists i: 0 \leq i \leq \#xs : \text{sum}(xs \uparrow i) = 8 \rangle$$

$$\textcircled{CB} \text{sum8}.C = \langle \exists i: 0 \leq i \leq \#C : \text{sum}(C \uparrow i) = 8 \rangle$$

$$= \{ \text{Rango unitario} \}$$

$$\text{sum}(C \uparrow 0) = 8$$

$$= \{ \text{Def take} \}$$

$$\text{sum}.C = 8$$

$$= \{ \text{Def sum, logica} \}$$

$$\boxed{\text{False}}$$

$$\textcircled{CA} \text{sum8}.(x \mathbin{\text{D}} xs) = \langle \exists i: 0 \leq i \leq \#(x \mathbin{\text{D}} xs) : \text{sum}((x \mathbin{\text{D}} xs) \uparrow i) = 8 \rangle \quad (H)$$

$$= \{ \text{logica, Partición de rango} \}$$

$$\langle \exists i: i=0 : \text{sum}(x \mathbin{\text{D}} xs) \uparrow i = 8 \rangle \vee \langle \exists i: 1 \leq i \leq \#(x \mathbin{\text{D}} xs) : \text{sum}(x \mathbin{\text{D}} xs) \uparrow i = 8 \rangle$$

$$= \{ \text{cambio de variable } i := i+1, \text{aritmética, Def length} \}$$

$$\langle \exists i: 0 \leq i < \#xs : \text{sum}((x \mathbin{\text{D}} xs) \uparrow (i+1)) = 8 \rangle$$

$$= \{ \text{Def Take} \}$$

$$\langle \exists i: 0 \leq i < \#xs : \text{sum}(x \mathbin{\text{D}} (xs \uparrow i)) = 8 \rangle$$

$$= \{ \text{Def sum} \}$$

$$\langle \exists i: 0 \leq i \leq \#xs : x + \text{sum}(xs \uparrow i) = 8 \rangle$$

$$= \{ \text{Generalizo} \}$$

$$g\text{sum8}.n.xs = \langle \exists i: 0 \leq i \leq \#xs : n + \text{sum}(xs \uparrow i) = 8 \rangle \quad (H)$$

$$\textcircled{CB} g\text{sum8}.n.C = \langle \exists i: 0 \leq i \leq \#C : n + \text{sum}(C \uparrow i) = 8 \rangle$$

$$= \{ \text{Def length, Rango unitario} \}$$

$$n + \text{sum}(C \uparrow 0) = 8$$

$$= \{ \text{Def take, Def sum} \}$$

$$\boxed{n=8}$$

$$\textcircled{CA} g\text{sum8}.n.(x \mathbin{\text{D}} xs) = \langle \exists i: 0 \leq i \leq \#(x \mathbin{\text{D}} xs) : n + \text{sum}((x \mathbin{\text{D}} xs) \uparrow i) = 8 \rangle$$

$$= \{ \text{logica, Partición de rango} \}$$

$$\langle \exists i: i=0 : n + \text{sum}((x \mathbin{\text{D}} xs) \uparrow i) = 8 \rangle \vee \langle \exists i: 1 \leq i \leq \#(x \mathbin{\text{D}} xs) : n + \text{sum}((x \mathbin{\text{D}} xs) \uparrow i) = 8 \rangle$$

$$= \{ \text{Rango unitario} \}$$

$$n + \text{sum}((x \mathbin{\text{D}} xs) \uparrow 0) = 8 \vee \langle \exists i: 1 \leq i \leq \#(x \mathbin{\text{D}} xs) : n + \text{sum}((x \mathbin{\text{D}} xs) \uparrow i) = 8 \rangle$$

$$= \{ \text{cambio de variable } i = i+1, \text{aritmética, Def length} \}$$

$$n + \text{sum}((x \mathbin{\text{D}} xs) \uparrow 0) = 8 \vee \langle \exists i: 0 \leq i < \#xs : n + \text{sum}((x \mathbin{\text{D}} xs) \uparrow (i+1)) = 8 \rangle$$

$$= \{ \text{Def de take} \}$$

$$n + \text{sum}.C = 8 \vee \langle \exists i: 0 \leq i < \#xs : n + \text{sum}(x \mathbin{\text{D}} (xs \uparrow i)) = 8 \rangle$$

$$= \{ \text{Def sum} \}$$

$$n = 8 \vee \langle \exists i: 0 \leq i < \#xs : (n+x) + \text{sum}.(xs \uparrow i) = 8 \rangle$$

$$= \{ \text{HI} \}$$

$$n=8 \vee g\text{sum8}.(n+x).xs$$

$$g\text{sum8}: \text{Num} \rightarrow [\text{Num}] \rightarrow \text{Bool}$$

$$g\text{sum8}.n.C \equiv (n=8)$$

$$g\text{sum8}.n.(x \mathbin{\text{D}} xs) \equiv (n=8) \vee g\text{sum8}.(n+x).xs$$

d) $F.xs \equiv \langle \text{Max}.i: 0 \leq i < \#xs \wedge \text{sum}(xs \uparrow i) = \text{sum}(xs \downarrow i): i \rangle$

Ⓢ $F.CJ \equiv \langle \text{Max}.i: 0 \leq i < \#CJ \wedge \text{sum}(CJ \uparrow i) = \text{sum}(CJ \downarrow i): i \rangle$

= {Def length, leibniz, take, Drop}

$\langle \text{Max}.i: i=0 \wedge \text{sum} CJ = \text{sum} CJ: i \rangle$

= {Rango unitario}

Ⓢ, Ⓢ $F.(xDxs) \equiv \langle \text{Max}.i: 0 \leq i < \#(xDxs) \wedge \text{sum}(xs \uparrow i) = \text{sum}(xs \downarrow i): i \rangle$

= {logica, distributividad, conv, Partición de rango}

$\langle \text{Max}.i: i=0 \wedge \text{sum}(xs \uparrow i) = \text{sum}(xs \downarrow i): i \rangle \max \langle \text{Max}.i: 1 \leq i < \#(xDxs) \wedge \text{sum}(xs \uparrow i) = \text{sum}(xs \downarrow i): i \rangle$

= {logica, Cambio variable $i := i + 1$ aritmética, Def length}

$\langle \text{Max}.i: 0 \leq i < \#xs \wedge \text{sum}(\cancel{xDxs} \uparrow (i+1)) = \text{sum}(\cancel{xDxs} \downarrow (i+1)): (i+1) \rangle$

= {Def take, Def Drop}

$\langle \text{Max}.i: 0 \leq i < \#xs \wedge \text{sum}(xD(xs \uparrow i)) = \text{sum}(xD(xs \downarrow i)): (i+1) \rangle$

= {Def Sum}

$\langle \text{Max}.i: 0 \leq i < \#xs \wedge \underline{x + \text{sum}(xs \uparrow i)} = \underline{x + \text{sum}(xs \downarrow i)}: (i+1) \rangle$

= {generalizo}

gf.n.xs $\equiv \langle \text{Max}.i: 0 \leq i < \#xs \wedge x + \text{sum}(xs \uparrow i) = x + \text{sum}(xs \downarrow i): i \rangle + 1$

Ⓢ gf.n.CJ $\equiv \langle \text{Max}.i: 0 \leq i < \#CJ \wedge x + \text{sum}(CJ \uparrow i) = x + \text{sum}(CJ \downarrow i): i \rangle$

= {Def length, Rango unitario}

$\langle \text{Max}.i: i=0 \wedge x + \text{sum}(CJ \uparrow i) = x + \text{sum}(CJ \downarrow i): i \rangle$

= {leibniz, Def take, Def Drop}

$\langle \text{Max}.i: i=0 \wedge x + \text{sum} CJ = x + \text{sum} CJ: 0 \rangle$

= {Rango unitario}

Ⓢ

Ⓢ gf.n.(xD(yDxs)) $\equiv \langle \text{Max}.i: 0 \leq i < \#(xD(yDxs)) \wedge \text{sum}((xD(yDxs)) \uparrow i) = x + \text{sum}((xD(yDxs)) \downarrow i): i \rangle$

= {logica, distributividad, Partición de rango}

$\langle \text{Max}.i: i=0 \wedge x + \text{sum}((xD(yDxs)) \uparrow i) = x + \text{sum}((xD(yDxs)) \downarrow i): i \rangle \max \langle \text{Max}.i: 1 \leq i < \#(xD(yDxs)) \wedge$

= {leibniz}

$\langle \text{Max}.i: i=0 \wedge x + \text{sum} CJ = x + \text{sum} CJ: 0 \rangle \max \langle \text{Max}.i: 1 \leq i < \#(xD(yDxs)) \wedge$

= {Rango unitario}

$0 \max \langle \text{Max}.i: 1 \leq i < \#(xD(yDxs)) \wedge x + \text{sum}((xD(yDxs)) \uparrow i) = x + \text{sum}((xD(yDxs)) \downarrow i): i \rangle$

= {Def length, Cambio de Variable, aritmética?}

$$0 \text{ max } < \text{Max } i: 0 \leq i < \#(x \upharpoonright x_s) \wedge x + \text{sum}(x \upharpoonright (x \upharpoonright x_s) \upharpoonright i) = x + \text{sum}(x \upharpoonright (x \upharpoonright x_s) \downarrow i) : (i+1) >$$

= {Def take, Def Drop?}

$$0 \text{ max } < \text{Max } i: 0 \leq i < \#(x \upharpoonright x_s) \wedge x + \text{sum}(x \upharpoonright (x \upharpoonright x_s) \upharpoonright i) = x + \text{sum}(x \upharpoonright (x \upharpoonright x_s) \downarrow i) : (i+1) >$$

= {Def sum, termino constante?}

$$0 \text{ max } < \text{Max } i: 0 \leq i < \#(x \upharpoonright x_s) \wedge (n+x) + \text{sum}(x \upharpoonright x_s \upharpoonright i) = (x+n) + \text{sum}(x \upharpoonright x_s \downarrow i) : i > + 1$$

= {HI?}

$$0 \text{ max } \text{gf. } (x+n) \cdot (x \upharpoonright x_s) + 2 \quad ??$$

✓ 13. Expresar utilizando cuantificadores las siguientes sentencias del lenguaje natural:

- a) La lista xs es un segmento inicial de la lista ys .

$$a) < \exists x_s: y_s = a_s ++ b_s : x_s = a_s >$$

- b) La lista xs es un segmento de la lista ys .

$$c) < \exists x_s: y_s = a_s ++ b_s : x_s = b_s >$$

c) La lista xs es un segmento final de la lista ys .

$$b) < \exists x_s: y_s = a_s ++ b_s ++ c_s : x_s = b_s >$$

14. Derivar funciones recursivas a partir de cada una de las especificaciones que escribiste para los ejercicios 13a y 13b.

$$a) \text{ seg-In. } x_s. y_s = < \exists a_s, b_s: y_s = a_s ++ b_s : x_s = a_s > \textcircled{H}$$

$$\textcircled{C} \text{ seg-In. } [] . y_s = < \exists a_s, b_s: y_s = a_s ++ b_s : [] = a_s >$$

= {lógica?}

$$< \exists a_s, b_s: y_s = a_s ++ b_s \wedge (a_s = [] \vee a_s \neq []) : [] = a_s >$$

= {Distributiva y Partición de rango?}

$$< \exists a_s, b_s: y_s = a_s ++ b_s \wedge a_s = [] : [] = a_s > \vee < \exists a_s, b_s: y_s = a_s ++ b_s \wedge a_s \neq [] : [] = a_s >$$

= {eliminación de variable?}

$$< \exists a_s, b_s: y_s = [] ++ b_s : [] = [] > \vee < \exists a_s, b_s: y_s = a_s ++ b_s \wedge a_s \neq [] : [] = a_s >$$

= {Def concatenar?}

$$< \exists a_s, b_s: y_s = b_s : [] = [] > \vee < \exists a_s, b_s: y_s = a_s ++ b_s \wedge a_s \neq [] : [] = a_s >$$

= {termino constante, lógica?}

$$\text{True} \vee < \exists a_s, b_s: y_s = a_s ++ b_s \wedge a_s \neq [] : [] = a_s >$$

= {cambio de variable $as := (a \triangleright as)$ }

$$true \vee \langle \exists as, bs: ys = (a \triangleright as) \uparrow \uparrow bs : \underline{C} \rangle = (a \triangleright as) \rangle$$

= {termino constante}

$$true \vee false = \underline{1 + true} //$$

$$(CB2) \text{ seg-in. } xs : C \rangle = \langle \exists as, bs : \underline{C} \rangle = as \uparrow \uparrow bs : xs = as \rangle$$

= {Prop Concatenación}

$$\langle \exists as, bs : \underline{as = C} \wedge \underline{bs = C} : xs = as \rangle$$

= {eliminación de variable, término constante}

$$\underline{1 + xs = C} //$$

$$(CR) \text{ seg-in. } (x \triangleright xs), (y \triangleright ys) = \langle \exists as, bs : \underline{(y \triangleright ys) = as \uparrow \uparrow bs} : (x \triangleright xs) = as \rangle$$

= {lógica, distributividad, Partición de rango}

$$\langle \exists as, bs : (y \triangleright ys) = as \uparrow \uparrow bs \wedge \underline{as = C} : (x \triangleright xs) = as \rangle \vee \langle \exists as, bs : (y \triangleright ys) = as \uparrow \uparrow bs \wedge \underline{as \neq C} : (x \triangleright xs) = as \rangle$$

= {eliminación de variable}

$$\langle \exists as, bs : (y \triangleright ys) = \underline{C} \uparrow \uparrow bs : (x \triangleright xs) = \underline{C} \rangle \vee \langle \exists as, bs : (y \triangleright ys) = as \uparrow \uparrow bs \wedge \underline{as \neq C} : (x \triangleright xs) = as \rangle$$

= {termino constante}

$$(x \triangleright xs) = \underline{C} \vee \langle \exists as, bs : (y \triangleright ys) = as \uparrow \uparrow bs \wedge \underline{as \neq C} : (x \triangleright xs) = as \rangle$$

= {lógica (a $\triangleright$ as), cambio de var.}

$$(x \triangleright xs) = \underline{C} \vee \langle \exists as, bs : (y \triangleright ys) = \underline{(a \triangleright as) \uparrow \uparrow bs} : (x \triangleright xs) = (a \triangleright as) \rangle$$

= {Def concatenar}

$$(x \triangleright xs) = \underline{C} \vee \langle \exists as, bs : \underline{(y \triangleright ys) = (a \triangleright (as \uparrow \uparrow bs))} : (x \triangleright xs) = (a \triangleright as) \rangle$$

= {Prop de constructor}

$$(x \triangleright xs) = \underline{C} \vee \langle \exists as, bs : \underline{y = a} \wedge \underline{ys = (as \uparrow \uparrow bs)} : (x \triangleright xs) = (a \triangleright as) \rangle$$

= {eliminación de variable}

$$(x \triangleright xs) = \underline{C} \vee \langle \exists as, bs : \underline{ys = (as \uparrow \uparrow bs)} : \underline{(x \triangleright xs) = (a \triangleright as)} \rangle$$

= {Prop constructor}

$$(x \triangleright xs) = \underline{C} \vee \langle \exists as, bs : \underline{ys = (as \uparrow \uparrow bs)} : \underline{x = y} \wedge \underline{xs = as} \rangle$$

= {Distributividad}

$$(x \triangleright xs) = \underline{C} \vee \langle \exists as, bs : \underline{ys = (as \uparrow \uparrow bs)} : \underline{xs = as} \rangle \wedge \underline{x = y}$$

$$= \{HI\} \quad \underline{(x \triangleright xs) = C} \vee \text{seg-in. } xs. ys \wedge x = y$$

$$\text{seg-in. } [a] \rightarrow [a] \rightarrow \text{Bool}$$

$$\text{seg-in. } C. ys = \text{true}$$

$$\text{seg-in. } xs : C \rangle = (xs = C)$$

$$\text{seg-in. } (x \triangleright xs), (y \triangleright ys) = \text{seg-in. } xs, ys \wedge x = y$$

= {Prop listas, absorbente} Seg. in. xs. ys ^ x = y

b) Seg. xs. ys = $\langle \exists a_s, b_s, c_s: ys = a_s ++ b_s ++ c_s : xs = bs \rangle$ ^(H)

CB Seg. C. ys = $\langle \exists a_s, b_s, c_s: ys = a_s ++ b_s ++ c_s : C \rangle = bs \rangle$

= {lógica, distributividad, partición de rango?}

$\langle \exists a_s, b_s, c_s: ys = a_s ++ b_s ++ c_s \wedge bs = C : C = bs \rangle \vee \langle \exists a_s, b_s, c_s: ys = a_s ++ b_s ++ c_s \wedge bs \neq C : C = bs \rangle$

= {eliminación de variable}

$\langle \exists a_s, b_s, c_s: ys = a_s ++ C ++ c_s : C = C \rangle \vee \langle \exists a_s, b_s, c_s: ys = a_s ++ b_s ++ c_s \wedge bs \neq C : C = bs \rangle$

= {termino constante?}
true $\vee \langle \exists a_s, b_s, c_s: ys = a_s ++ b_s ++ c_s \wedge bs \neq C : C = bs \rangle$

= {Cambio de variable, bs := (b ++ bs), termino constante}

true $\vee$ False = True

CB2 Seg. xs. C = $\langle \exists a_s, b_s, c_s: C = a_s ++ b_s ++ c_s : xs = bs \rangle$

= {Prop listas, concatenar}

$\langle \exists a_s, b_s, c_s: a_s = C \wedge b_s = C \wedge c_s = C : xs = C \rangle$

= {termino constante}

xs = C

CR Seg. (x ++ xs). (y ++ ys) = $\langle \exists a_s, b_s, c_s: (y ++ xs) = a_s ++ b_s ++ c_s : (x ++ xs) = bs \rangle$

= {lógica, distributividad, partición de rango}

$\langle \exists a_s, b_s, c_s: (y ++ xs) = a_s ++ b_s ++ c_s \wedge a_s = C : (x ++ xs) = bs \rangle \vee \langle \exists a_s, b_s, c_s: (y ++ xs) = a_s ++ b_s ++ c_s \wedge a_s \neq C : (x ++ xs) = bs \rangle$

= {eliminación de variable}

$\langle \exists a_s, b_s, c_s: (y ++ xs) = C ++ b_s ++ c_s : (x ++ xs) = bs \rangle \vee \langle \exists a_s, b_s, c_s: (y ++ xs) = a_s ++ b_s ++ c_s \wedge a_s \neq C : (x ++ xs) = bs \rangle$

= {Prop concatenar}

$\langle \exists a_s, b_s, c_s: (y ++ xs) = b_s ++ c_s : (x ++ xs) = bs \rangle \vee \langle \exists a_s, b_s, c_s: (y ++ xs) = a_s ++ b_s ++ c_s \wedge a_s \neq C : (x ++ xs) = bs \rangle$

• {Modularizo}

mseg. ys = $\langle \exists a_s, b_s, c_s: ys = b_s ++ c_s : (x ++ xs) = bs \rangle$ ^(H)

15. Derivar las funciones especificadas como:

- a) $\text{sumin}.xs = \langle \text{Min } as, bs, cs : xs = as ++ bs ++ cs : \text{sum}.bs \rangle$
(suma mínima de un segmento).
- b) $f.xs = \langle N as, bs, cs : xs = as ++ bs ++ cs : 8 = \text{sum}.bs \rangle$
(cantidad de segmentos que suman eso).
- c) $\text{maxiga.e}.xs = \langle \text{Max } as, bs, cs : xs = as ++ bs ++ cs \wedge \text{iga.e}.bs : \#bs \rangle$
(máxima longitud de iguales a e)
donde *iga* es la función del ejercicio 3b.

$$\text{min}.xs = \langle \text{Min } as, bs, cs : xs = as ++ bs ++ cs : \text{sum}.bs \rangle \quad \textcircled{H}$$

$$\textcircled{C} \text{ min}.(x \text{D} xs) \equiv \langle \text{Min } as, bs, cs : (x \text{D} xs) = as ++ bs ++ cs : \text{sum}.bs \rangle$$

= {lógica, distributiva, Partición de rango}

$$\langle \text{Min } as, bs, cs : (x \text{D} xs) = as ++ bs ++ cs \wedge \underline{as = []} : \text{sum}.bs \rangle \text{ min } \langle \text{Min } as, bs, cs : (x \text{D} xs) = as ++ bs ++ cs \wedge \underline{as \neq []} : \text{sum}.bs \rangle$$

= {eliminación de variable}

$$\langle \text{Min } as, bs, cs : (x \text{D} xs) = \underline{[]} ++ bs ++ cs : \text{sum}.bs \rangle \text{ min } \langle \text{Min } as, bs, cs : (x \text{D} xs) = as ++ bs ++ cs \wedge \underline{as \neq []} : \text{sum}.bs \rangle$$

= {Def concatenar}

$$\langle \text{Min } as, bs, cs : (x \text{D} xs) = bs ++ cs : \text{sum}.bs \rangle \quad \text{min } \langle \text{Min } as, bs, cs : (x \text{D} xs) = as ++ bs ++ cs \wedge \underline{as \neq []} : \text{sum}.bs \rangle$$

= {modularizo}

Especificación

$$\text{Smin}.xs = \langle \text{Min } as, bs, cs : xs = bs ++ cs : \text{sum}.bs \rangle \quad \textcircled{H}$$

$$\textcircled{CB} \text{ Smin } [] \equiv \langle \text{Min } as, bs, cs : \underline{[]} = bs ++ cs : \text{sum}.bs \rangle$$

= {Def concatenar}

$$\langle \text{Min. } a_s, b_s, c_s : \underline{b_s = C} \wedge c_s = C : \text{sum. } b_s \rangle \quad \textcircled{CR} \text{ smin.}(x \triangleright x_s) = \langle \text{Min. } a_s, b_s, c_s : \underline{(x \triangleright x_s) = b_s + c_s} : \text{sum. } b_s \rangle$$

= {eliminación de variable c_s , Rango unitario} = {lógica}

$$\text{sum. } C \quad \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = b_s + c_s \wedge (b_s = C \vee b_s \neq C) : \text{sum. } b_s \rangle$$

= {Def de sum}

= {Distributividad del $\wedge$ con el $\vee$, Partición de rango}

$$\boxed{\text{O}} \quad \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = b_s + c_s \wedge \underline{b_s = C} : \text{sum. } b_s \rangle \min \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = b_s + c_s \wedge \underline{b_s \neq C} : \text{sum. } b_s \rangle$$

= {eliminación de variable b_s }

$$\langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = C + c_s : \text{sum. } C \rangle \min \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = b_s + c_s \wedge \underline{b_s \neq C} : \text{sum. } b_s \rangle$$

= {Def de sum, término constante}

$$\text{O} \min \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = \underline{b_s + c_s} \wedge \underline{b_s \neq C} : \text{sum. } b_s \rangle$$

= {cambio de variable, $b_s := (b \triangleright b)$ }

$$\text{O} \min \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = (b \triangleright b) + c_s : \text{sum. } (b \triangleright b) \rangle$$

= {Def concatenar}

$$\text{O} \min \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = \underline{b \triangleright (b_s + c_s)} : \text{sum. } (b \triangleright b_s) \rangle$$

= {Prop. constructor}

$$\text{O} \min \langle \text{Min. } a_s, b_s, c_s : \underline{x = b} \wedge x_s = b_s + c_s : \text{sum. } (b \triangleright b_s) \rangle$$

= {eliminación de variable, Definición de sum}

$$\text{O} \min \langle \text{Min. } a_s, b_s, c_s : x_s = b_s + c_s : \underline{x + \text{sum. } b_s} \rangle$$

= {Distributividad}

$$\text{O} \min (\langle \text{Min. } a_s, b_s, c_s : x_s = b_s + c_s : \text{sum. } b_s \rangle + x)$$

= {HI}

$$\text{O} \min (\text{smin. } x_s + x) \min \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = a_s + b_s + c_s \wedge \underline{a_s \neq C} : \text{sum. } b_s \rangle$$

= {cambio de variable, $a_s := (a \triangleright a_s)$ }

$$\text{O} \min (\text{smin. } x_s + x) \min \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = \underline{(a \triangleright a_s) + b_s + c_s} : \text{sum. } b_s \rangle$$

= {Def concatenar}

$$\text{O} \min (\text{smin. } x_s + x) \min \langle \text{Min. } a_s, b_s, c_s : (x \triangleright x_s) = a \triangleright (a_s + b_s + c_s) : \text{sum. } b_s \rangle$$

= {Prop del constructor}

$$\text{O} \min (\text{smin. } x_s + x) \min \langle \text{Min. } a_s, b_s, c_s : \underline{x = a} \wedge x_s = a_s + b_s + c_s : \text{sum. } b_s \rangle$$

= {eliminación de variable}

$$\text{O} \min (\text{smin. } x_s + x) \min \langle \text{Min. } a_s, b_s, c_s : x_s = a_s + b_s + c_s : \text{sum. } b_s \rangle$$

= {HI}

$$\text{O} \min (\text{smin. } x_s + x) \min \min. x_s$$

$$(C) \min.C = \langle \text{Min } a_s, b_s, c_s : C = a_s + b_s + c_s : \text{sum.bs} \rangle$$

= {Prop de concatenar}

$$\langle \text{Min. } a_s, b_s, c_s : a_s = C \wedge b_s = C \wedge c_s = C : \text{sum.bs} \rangle$$

= {eliminación de variables, Rango unitario, Def suma}

$[0]_4$

$$\therefore \min: [Num] \rightarrow Num$$

$$\min.C = 0$$

$$\min.(x \text{D} x_s) = \min(x \text{D} x_s) \min \min.x_s$$

$$S\min: [Num] \rightarrow Num$$

$$S\min.C = 0$$

$$S\min.(x \text{D} x_s) = 0 \min (S\min.x_s + x)$$

$$b) f.xs = \langle N a_s, b_s, c_s : xs = a_s + b_s + c_s : 8 = \text{sum.bs} \rangle$$

(cantidad de segmentos que suman eso).

$$f.xs = \langle \sum a_s, b_s, c_s : x = a_s + b_s + c_s \wedge 8 = \text{sum.bs} : 1 \rangle \quad (H)$$

$$(C) f.(x \text{D} x_s) = \langle \sum a_s, b_s, c_s : (x \text{D} x_s) = a_s + b_s + c_s \wedge 8 = \text{sum.bs} : 1 \rangle$$

= {lógica, Partición de rango}

$$\langle \sum a_s, b_s, c_s : (x \text{D} x_s) = (a_s + b_s + c_s \wedge a_s = C) \wedge 8 = \text{sum.bs} : 1 \rangle + \langle \sum a_s, b_s, c_s : (x \text{D} x_s) = (a_s + b_s + c_s \wedge a_s \neq C) \wedge 8 = \text{sum.bs} : 1 \rangle$$

= {eliminación de variable, Def concat}

$$\langle \sum b_s, c_s : (x \text{D} x_s) = b_s + c_s \wedge 8 = \text{sum.bs} : 1 \rangle + \langle \sum a_s, b_s, c_s : (x \text{D} x_s) = (a_s + b_s + c_s \wedge a_s \neq C) \wedge 8 = \text{sum.bs} : 1 \rangle$$

(H)
= {Modularización}

$$mf.xs = \langle \sum b_s, c_s : x \text{D} x_s = b_s + c_s \wedge 8 = \text{sum.bs} : 1 \rangle \quad (H)$$

$$(C) mf.C = \langle \sum b_s, c_s : C = b_s + c_s \wedge 8 = \text{sum.bs} : 1 \rangle$$

= {Prop de listas}

$$\langle \sum b_s, c_s : b_s = C \wedge c_s = C \wedge 8 = \text{sum.bs} : 1 \rangle$$

= {eliminación de variable (C), leibniz}

$$\langle \sum b_s, c_s : b_s = C \wedge 8 = \text{sum.C} : 1 \rangle$$

= {Def de sum, absorbente del 1}

$$\langle \sum b_s, c_s : \text{false} : 1 \rangle$$

= {Rango vacío}

$[0]_4$

$$\textcircled{1} \text{ m.f.}(x \bowtie x) = \langle \sum b_i, c_i : (x \bowtie x) = b_i + c_i \wedge 8 = \text{sum. } b_i : 1 \rangle$$

= {lógica, Distributividad, Partición de rango}

$$\langle \sum b_i, c_i : (x \bowtie x) = (b_i + c_i \wedge b_i = C) \wedge 8 = \text{sum. } b_i : 1 \rangle + \langle \sum b_i, c_i : (x \bowtie x) = (b_i + c_i \wedge b_i \neq C) \wedge 8 = \text{sum. } b_i : 1 \rangle$$

= {eliminación de variable, def concat}

$$\langle \sum c_i : (x \bowtie x) = b_i \wedge 8 = \text{sum } C : 1 \rangle + \langle \sum b_i, c_i : (x \bowtie x) = (b_i + c_i \wedge b_i \neq C) \wedge 8 = \text{sum. } b_i : 1 \rangle$$

= {Def sum, lógica, absorbente del $\wedge$, Rango vacío}

$$0 + \langle \sum b_i, c_i : (x \bowtie x) = (b_i + c_i \wedge b_i \neq C) \wedge 8 = \text{sum. } b_i : 1 \rangle$$

= {Cambio de variable. $b_i := (b_i \bowtie b_i)$ }

$$\langle \sum b_i, c_i : (x \bowtie x) = (b_i \bowtie b_i) + c_i \wedge 8 = \text{sum. } (b_i \bowtie b_i) : 1 \rangle$$

= {Def Concatenar, Prop listas}

$$\langle \sum b_i, c_i : x = b_i \wedge x = b_i + c_i \wedge 8 = \text{sum. } (b_i \bowtie b_i) : 1 \rangle$$

= {eliminación de variable, Def sum}

$$\langle \sum b_i, c_i : x = b_i + c_i \wedge 8 = x + \text{sum. } b_i : 1 \rangle$$

= {generalización}

$$\text{gm.f. } n. x = \langle \sum b_i, c_i : x = b_i + c_i \wedge 8 = n + \text{sum. } b_i : 1 \rangle$$

$\textcircled{2}$

$$\text{gm.f. } n. C = \langle \sum b_i, c_i : C = b_i + c_i \wedge 8 = n + \text{sum. } b_i : 1 \rangle$$

= {Prop listas, eliminación de var, Def sum}

$$\langle \sum b_i, c_i : 8 = n : 1 \rangle$$

= {termino constante}

$$\textcircled{1} \text{ m.f. } n. (x \bowtie x) = \langle \sum b_i, c_i : (x \bowtie x) = b_i + c_i \wedge 8 = n + \text{sum. } b_i : 1 \rangle$$

$\boxed{1}$

= {lógica distributiva, Partición de rango}

$$\langle \sum b_i, c_i : (x \bowtie x) = b_i + c_i \wedge b_i = C \wedge 8 = n + \text{sum. } b_i : 1 \rangle + \langle \sum b_i, c_i : (x \bowtie x) = b_i + c_i \wedge b_i \neq C \wedge 8 = n + \text{sum. } b_i : 1 \rangle$$

= {eliminación de variable, Def concat, Def suma}

$$\langle \sum b_i, c_i : (x \bowtie x) = c_i \wedge 8 = n : 1 \rangle$$

= {termino constante}

$$1 + \langle \sum b_i, c_i : (x \bowtie x) = b_i + c_i \wedge b_i \neq C \wedge 8 = n + \text{sum. } b_i : 1 \rangle$$

= {Cambio de variable $b_i := (b_i \bowtie b_i)$, Def concat, Prop listas, Def sum}

$$1 + \langle \sum b_i, c_i : x = b_i + c_i \wedge 8 = (n+x) + \text{sum } b_i : 1 \rangle$$

= {HD}

$$1 + \text{gm.f. } (n+x). x$$

$$1 + \text{gmf}.(n+x).xs + \langle \sum a_i, b_i, c_i : (x \text{D} x_i) = (a_i \text{++} b_i \text{++} c_i \wedge a_i \neq c_i) \wedge \# = \text{sum } b_i : 1 \rangle$$

= { Cambio de variable $b_i := (b_i \text{D} b_i)$, Def Concat, Prop listas, Def sum }

$$1 + \text{gmf}.(n+x).xs + \langle \sum a_i, b_i, c_i : x_i = (a_i \text{++} b_i \text{++} c_i) \wedge \# = \text{sum } b_i : 1 \rangle$$

= { HI }

$$1 + \text{gmf}.(n+x).xs + \text{f}.xs$$

ⓐ

$$\text{f}.[] = \langle \sum a_i, b_i, c_i : [] = a_i \text{++} b_i \text{++} c_i \wedge \# = \text{sum } b_i : 1 \rangle$$

= { todas las aplicaciones. }

ⓑ

$$\text{f} : [\text{Num}] \rightarrow \text{Num}$$

$$\text{f}[] = 0$$

$$\text{f}(x \text{D} x_i) = 1 + \text{gmf}.x.xs + \text{f}.x_i$$

$$\text{gmf} : \text{Num} \rightarrow [\text{Num}] \rightarrow \text{Num}$$

$$\text{gmf}.n.[] = 1$$

$$\text{gmf}.n.(x \text{D} x_i) = 1 + \text{gmf}.(n+x).x_i$$

$$\text{c) } \text{maxiga.e.xs} = \langle \text{Max } a_i, b_i, c_i : xs = a_i \text{++} b_i \text{++} c_i \wedge \text{iga.e.bs} : \#bs \rangle$$

máxima longitud de iguales a e)

donde iga es la función del ejercicio 3b.

ⓓ

$$\text{maxiga.e.xs} = \langle \text{Max } a_i, b_i, c_i : xs = a_i \text{++} b_i \text{++} c_i \wedge \text{iga.e.bs} : \#bs \rangle$$

$$\text{ⓔ) } \text{max.e}.[] = \langle \text{Max } a_i, b_i, c_i : [] = a_i \text{++} b_i \text{++} c_i \wedge \text{iga.e.bs} : \#bs \rangle$$

= { Prop concat, eliminación de variable, Def iga }

$$\langle \text{Max } bs : \text{true} : \#[] \rangle$$

= { Def length }

$$\text{ⓕ) } \text{Max.e.}(x \text{D} x_i) = \langle \text{Max } a_i, b_i, c_i : (x \text{D} x_i) = a_i \text{++} b_i \text{++} c_i \wedge \text{iga.e.bs} : \#bs \rangle$$

ⓖ

= { lógica, Distributiva, Partición de rango }

$$\langle \text{Max } a_i, b_i, c_i : (x \text{D} x_i) = (a_i \text{++} b_i \text{++} c_i \wedge a_i = c_i) \wedge \text{iga.e.bs} : \#bs \rangle \overset{\text{max}}{\langle \text{Max } a_i, b_i, c_i : (x \text{D} x_i) = (a_i \text{++} b_i \text{++} c_i \wedge a_i \neq c_i) \wedge \text{iga.e.bs} : \#bs \rangle}$$

= { eliminación de variable, Prop concat }

$$\langle \text{Max } b_i, c_i : (x \text{D} x_i) = b_i \text{++} c_i \wedge \text{iga.e.bs} : \#bs \rangle \text{max} \langle \text{Max } a_i, b_i, c_i : (x \text{D} x_i) = (a_i \text{++} b_i \text{++} c_i \wedge a_i \neq c_i) \wedge \text{iga.e.bs} : \#bs \rangle$$

= { modularización }

$$\text{Mmaxiga.e.xs} = \langle \text{Max } b_i, c_i : xs = b_i \text{++} c_i \wedge \text{iga.e.bs} : \#bs \rangle$$

$$\text{ⓗ) } \text{Mmaxiga.e}.[] = \langle \text{Max } b_i, c_i : [] = b_i \text{++} c_i \wedge \text{iga.e.bs} : \#bs \rangle$$

= { Def concat, eliminación de var, Def iga }

$$\langle \text{Max } c_i : \text{true} : \#[] \rangle$$

= { Def length }

$$\text{ⓔ) } \text{Mmaxiga.e.}(x \text{D} x_i) = \langle \text{Max } b_i, c_i : (x \text{D} x_i) = b_i \text{++} c_i \wedge \text{iga.e.bs} : \#bs \rangle$$

ⓖ

= { lógica, Distributividad, Partición de rango }

$$\langle \text{Max } b_i, c_i : (x \text{D} x_i) = (b_i \text{++} c_i \wedge b_i = c_i) \wedge \text{iga.e.bs} : \#bs \rangle \overset{\text{max}}{\langle \text{Max } b_i, c_i : (x \text{D} x_i) = (b_i \text{++} c_i \wedge b_i \neq c_i) \wedge \text{iga.e.bs} : \#bs \rangle}$$

= { Prop concat, eliminación variable, Def length, término constante }

$$0 \text{ max } \langle \text{Max } b, c, s : (x \text{D} x s) = (b \text{++} c s \wedge b s \neq c) \wedge \text{iga.e.} b s : \# b s \rangle$$

= {Cambio de variable $b s := (b \text{D} b s)$, Def concat, Prop del constructor, Def length}

$$0 \text{ max } \langle \text{Max } b, c, s : \underline{x = b} \wedge x s = b \text{++} c s \wedge \text{iga.e.} (b \text{D} b s) : \# b s + 1 \rangle$$

= {eliminación de variable, def iga y distributiva}

$$0 \text{ max } \langle \text{Max } b, c, s : x s = b \text{++} c s \wedge (x = e) \wedge \text{iga.e.} b s : \# b s + 1 \rangle$$

= {eliminación de variable, HI}

$$0 \text{ max } (M \text{maxiga.e.} x s + 1)$$

$$0 \text{ max } (M \text{maxiga.e.} x s + 1) \text{ max } \langle \text{Max } a, b, c, s : (x \text{D} x s) = (a s \text{++} b s \text{++} c s \wedge a s \neq c) \wedge \text{iga.e.} b s : \# b s \rangle$$

= {Cambio de variable $a s := (a \text{D} a s)$, Def concat, Def constructor}

$$0 \text{ max } (M \text{maxiga.e.} x s + 1) \text{ max } \langle \text{Max } a, b, c, s : \underline{x = a} \wedge x s = a s \text{++} b s \text{++} c s \wedge \text{iga.e.} b s : \# b s \rangle$$

= {eliminación de variable, HI}

$$0 \text{ max } (M \text{maxiga.e.} x s + 1) \text{ max } \text{maxiga.e.} x s$$

$$\text{iga} : \text{Int} \rightarrow [\text{Int}] \rightarrow \text{Bool}$$

$$\text{iga.v.} [] = \text{true}$$

$$\text{iga.w} (x \text{D} x s) = (x = v) \wedge \text{iga.v.} x s$$

$$\text{maxiga} : \text{Num} \rightarrow [\text{Num}] \rightarrow \text{Num}$$

$$\text{maxiga.e.} [] = 0$$

$$\text{maxiga.e.} (x \text{D} x s) = (0 \text{ max } (M \text{maxiga.e.} x s + 1)) \text{ max } \text{maxiga.e.} x s$$

$$M \text{maxiga} : \text{Num} \rightarrow [\text{Num}] \rightarrow \text{Num}$$

$$M \text{maxiga.e.} [] = 0$$

$$M \text{maxiga.e.} (x \text{D} x s) = 0 \text{ max } (M \text{maxiga.e.} x s + 1)$$

16. Especificar los siguientes problemas:

- ✓ a) La lista $x s$ de números enteros tienen la misma cantidad de elementos pares e impares.
- ✓ b) Las listas $x s$ y $y s$ tienen en común un segmento (ojo que tiene trampa).
- c) La lista $x s$ posee un segmento **no** inicial y **no** final cuyos valores son mayores a los del resto de la misma.

$$a) \langle \sum i : 0 \leq i < \# x s, i \bmod 2 = 0 : 1 \rangle = \langle \sum j : 0 \leq j < \# x s, j \bmod 2 = 1 : 1 \rangle$$

$$b) \langle \exists a, b, s : x s = a s \text{++} b s : \langle \exists c, b, s : x s = c s \text{++} b s : a s = c s \rangle \rangle$$

$$c) \langle \exists a, b, c, s : x s = a s \text{++} (b \text{D} b s) \text{++} c s : b \wedge (b \text{D} b s) > x s \rangle$$

17. Especificar formalmente utilizando cuantificadores cada una de las siguientes funciones descriptas informalmente. Luego, *derivar* soluciones algorítmicas para cada una.

✓ a) $\text{cuad} : \text{Nat} \rightarrow \text{Bool}$, que dado un natural determina si es el cuadrado de un número.

✓ b) $\text{n8} : [\text{Num}] \rightarrow \text{Nat}$, que cuenta la cantidad de segmentos iniciales de una lista cuya suma es igual a 8.

$$a) \text{cuad.n} \langle \exists i : 1 \leq i \leq n : (i \cdot i) = n \rangle \text{ (H)}$$

$$\textcircled{CB} \text{Cvad. } 0 = \langle \exists i: 1 \leq i \leq n : (i \cdot i) = 0 \rangle$$

= {lógica, rango vacío}

$$\textcircled{CR} \text{Cvad. } (n+1) = \langle \exists i: 1 \leq i \leq (n+1) : (i \cdot i) = (n+1) \rangle$$

true

= {lógica, Partición de rango}

$$\langle \exists i: i = n+1 : (i \cdot i) = n+1 \rangle \vee \langle \exists i: 1 \leq i \leq n : (i \cdot i) = (n+1) \rangle$$

= {Rango unitario}

$$n^2 + 2n + 1 = n+1 \vee \langle \exists i: 1 \leq i \leq n : (i \cdot i) = (n+1) \rangle$$

$$n \cdot n + n = 0 \vee \langle \exists i: 1 \leq i \leq n : (i \cdot i) = (n+1) \rangle$$

= {generalización}

$$g\text{Cvad. } n.m = \langle \exists i: 1 \leq i \leq n : (i \cdot i) = n+m \rangle \quad \textcircled{H}$$

$$\textcircled{CB} g\text{Cvad. } 1.m = \langle \exists i: 1 \leq i \leq 1 : (i \cdot i) = m \rangle$$

= {lógica, Rango unit.}

m=0

$$\textcircled{CR} g\text{Cvad. } (n+1).m = \langle \exists i: 1 \leq i \leq n+1 : (i \cdot i) = (n+1)+m \rangle$$

= {lógica, Partición de rango}

$$\langle \exists i: i = n+1 : (i \cdot i) = (n+1)+m \rangle \vee \langle \exists i: 1 \leq i \leq n : (i \cdot i) = (n+1)+m \rangle$$

= {Rango unitario, aritmética}

$$n \cdot n + n = m \vee \langle \exists i: 1 \leq i \leq n : (i \cdot i) = (n+1)+m \rangle$$

= {asociatividad, HI}

$$n \cdot n + n = m \vee g\text{Cvad. } n.(n+1)$$

$$g\text{Cvad.} : \text{Num} \rightarrow \text{Num} \rightarrow \text{Bool}$$

$$g\text{Cvad. } 0.m = \text{true}$$

$$g\text{Cvad. } (n+1).m = n \cdot n + n = m \vee g\text{Cvad. } n.(n+1).$$

$$b) n8.xs = \langle \sum as, bs: xs = as ++ bs \wedge \text{sum}.as = 0 : 1 \rangle$$

$$\textcircled{CR} n8.(x \text{D} xs) = \langle \sum as, bs: (x \text{D} xs) = as ++ bs \wedge \text{sum}.as = 0 : 1 \rangle$$

= {lógica, Distributiva, Partición de rango}

$$\langle \sum as, bs: (x \text{D} xs) = (as ++ bs \wedge as = []) \wedge \text{sum}.as = 0 : 1 \rangle + \langle \sum as, bs: (x \text{D} xs) = (as ++ bs \wedge as \neq []) \wedge \text{sum}.as = 0 : 1 \rangle$$

= {eliminación de variable, Def concat, Def sum}

$$\langle \sum as, bs: (x \text{D} xs) = bs \wedge \text{false} : 1 \rangle + \langle \sum as, bs: (x \text{D} xs) = (as ++ bs \wedge as \neq []) \wedge \text{sum}.as = 0 : 1 \rangle$$

= {Absorbente del $\wedge$, Rango vacío}

$$0 + \langle \sum as, bs: (x \text{D} xs) = (as ++ bs \wedge as \neq []) \wedge \text{sum}.as = 0 : 1 \rangle$$

= {Cambio de variable $as \leftarrow (a \text{D} as)$, Def concat, Def sum, Prop del constructor}

$$\langle \Sigma a_i, b_i : x = a \wedge x_s = a_s + b_s \wedge (x + \text{sum } a_s = 0) : 1 \rangle$$

= {generalización}

$$\text{gnB}.xs.n = \langle \Sigma a_i, b_i : x_s = a_s + b_s \wedge (n + \text{sum } a_s = 0) : 1 \rangle \quad \textcircled{H}$$

$$\textcircled{CB} \text{gnB}[] . n = \langle \Sigma a_i, b_i : [] = a_s + b_s \wedge (n + \text{sum } a_s = 0) : 1 \rangle$$

= {Por concat, eliminación de variable b_i, Def sum}

$$\langle \Sigma a_i : n = 0 : 1 \rangle$$

1

$$\textcircled{CR} \text{ng}.(x \bowtie x_s) = \langle \Sigma a_i, b_i : (x \bowtie x_s) = a_s + b_s \wedge (n + \text{sum } a_s = 0) : 1 \rangle$$

= {lógica, Distributiva, Partición de rango}

$$\langle \Sigma a_i, b_i : (x \bowtie x_s) = a_s + b_s \wedge a_s = [] \wedge (n + \text{sum } a_s = 0) : 1 \rangle + \langle \Sigma a_i, b_i : (x \bowtie x_s) = a_s + b_s \wedge a_s \neq [] \wedge (n + \text{sum } a_s = 0) : 1 \rangle$$

= {eliminación de variable, Def concat, Def sum}

$$\langle \Sigma a_i, b_i : (x \bowtie x_s) = b_s \wedge (n = 0) : 1 \rangle$$

= {termino constante}

$$1 + \langle \Sigma a_i, b_i : (x \bowtie x_s) = a_s + b_s \wedge a_s \neq [] \wedge (n + \text{sum } a_s = 0) : 1 \rangle$$

{Cambio de variable $a_s \leftarrow (a \bowtie a_s)$, Def concat, Def sum, Por del constructor}

$$1 + \langle \Sigma a_i, b_i : x = a \wedge x_s = a_s + b_s \wedge ((n+x) + \text{sum } a_s = 0) : 1 \rangle$$

= {HI} $1 + \text{gnB}.xs.(n+x)$

= {Def conteo}

$$(n=0) \rightarrow 1 + \text{gnB}.xs.(n+x)$$

$$\neg(n=0) \rightarrow \text{gnB}.xs.(n+x)$$

)

$$\text{gnB} : [\text{Num}] \rightarrow \text{Num} \rightarrow \text{Num}$$

$$\text{gnB}[] . n = 1$$

$$\text{gnB}(x \bowtie x_s) . n \begin{cases} n=0 \rightarrow 1 + \text{gnB}.xs.(n+x) \\ \neg(n=0) \rightarrow \text{gnB}.xs.(n+x) \end{cases}$$

18. Describir en lenguaje natural y dar el tipo de cada una de las siguientes funciones especificadas formalmente:

a) $f.xs \doteq \langle N i, j : 0 \leq i < j < \#xs : xs.i = xs.j \rangle$

b) $g.xs \doteq \langle \text{Max } p, q : 0 \leq p < q < \#xs : xs.p + xs.q \rangle$

c) $h.xs \doteq \langle N k : 0 \leq k < \#xs : (\forall i : 0 \leq i < k : xs.i < xs.k) \rangle$

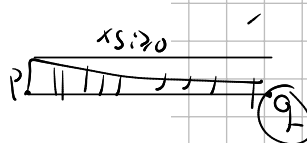
d) $k.xs \doteq \langle \forall i, j : 0 \leq i \wedge 0 \leq j \wedge i + j = \#xs - 1 : xs.i = xs.j \rangle$

e) $l.xs \doteq \langle \text{Max } p, q : 0 \leq p \leq q < \#xs \wedge (\forall i : p \leq i < q : xs.i \geq 0) : (q - p) \rangle$

19. Derivar la función definida en el ejercicio 18b.

a) $f : [\text{Num}] \rightarrow \text{Bool}$

"Todos los elementos de la lista xs son iguales."



b) $g: [Num] \rightarrow Int;$

"La suma entre los 2 elementos mas grandes de la lista."

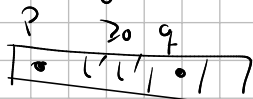
c) $h: [Num] \rightarrow Bool$

"La lista xs esta ordenada de forma creciente"

d) $k: [Num] \rightarrow Bool$

"la suma de las Posiciones i, j es igual tamaño de la lista, menos uno,

e) $t: [Num] \rightarrow Num$



" todos los elementos entre un máximo q y un máximo p son mayores que cero / se cumple $q-p$ "

19) b) $g.xs \doteq \langle Max p, q : 0 \leq p < q < \#xs : xs.p + xs.q \rangle$ (4)

20) $g.C \doteq \langle Max p, q : 0 \leq p < q < \#C : C.p + C.q \rangle$

$= \{ logica, Rango \text{ Varío} \}$

$[-p]$

21) $g.(x \triangleright xs) \doteq \langle Max p, q : 0 \leq p < q < \#(x \triangleright xs) : (x \triangleright xs).p + (x \triangleright xs).q \rangle$

$= \{ logica, Partición de Rango, Def length \}$

$\langle Max p, q : p=0 \wedge q=L : (x \triangleright xs).p + (x \triangleright xs).q \rangle \max \langle Max p, q : 1 \leq p < q < L + \#xs : (x \triangleright xs).p + (x \triangleright xs).q \rangle$

$= \{ Rango \text{ unitario, Def indexar} \}$

$(x + (x \triangleright xs).L) \max g.xs$

20. Derivar una definición recursiva para la función $f: [Num] \rightarrow [Num] \rightarrow Num$, que calcula la mínima distancia entre valores de las listas xs y ys , y cuya especificación es la siguiente:

$$f.xs.ys = \langle \text{Min } i, j : 0 \leq i < \#xs \wedge 0 \leq j < \#ys : |xs.i - ys.j| \rangle$$

(CB)

$$f.xs.ys = \langle \text{Min } i, j : 0 \leq i < \#xs \wedge 0 \leq j < \#ys : [xs.i - ys.j] \rangle^H$$

$$f.[] ys = \langle \text{Min } i, j : 0 \leq i < \#[] \wedge 0 \leq j < \#ys : [] . i - ys.j \rangle$$

$$= \{ \text{Def length, lógica, absorbente del } \wedge, \text{Rango vacío} \}$$

$$\boxed{1 + \infty}$$

$$f.xs.ys = \langle \text{Min } i, j : 0 \leq i < \#xs \wedge 0 \leq j < \#[] : [xs.i - [] . j] \rangle$$

$$= \{ \text{Def length, lógica, absorbente del } \wedge, \text{Rango vacío} \}$$

$$\boxed{1 + \infty}$$

$$f.(x \triangleright xs).(y \triangleright ys) = \langle \text{Min } i, j : 0 \leq i < \#(x \triangleright xs) \wedge 0 \leq j < \#(y \triangleright ys) : [(x \triangleright xs).i - (y \triangleright ys).j] \rangle$$

$$= \{ \text{lógica, Distributividad, Partición de rango} \}$$

$$\langle \text{Min } i, j : i = 0 \wedge j = 0 : [(x \triangleright xs).i - (y \triangleright ys).j] \rangle \min \langle \text{Min } i, j : 1 \leq i < \#(x \triangleright xs) \wedge 1 \leq j < \#(y \triangleright ys) : [(x \triangleright xs).i - (y \triangleright ys).j] \rangle$$

$$= \{ \text{Rango unitario, Def index} \}$$

$$[x - y] \min \langle \text{Min } i, j : 1 \leq i < \#(x \triangleright xs) \wedge 1 \leq j < \#(y \triangleright ys) : [(x \triangleright xs).i - (y \triangleright ys).j] \rangle$$

$$= \{ \text{Cambio de variable } i \leftarrow i + 1, j \leftarrow j + 1, \text{Def length, aritmética} \}$$

$$[x - y] \min \langle \text{Min } i, j : 0 \leq i < \#xs \wedge 0 \leq j < \#ys : [(x \triangleright xs).(i + 1) - (y \triangleright ys).(j + 1)] \rangle$$

$$= \{ \text{Def indexar, HI} \}$$

$$[x - y] \min f.xs.ys$$

$$f: [Num] \rightarrow [Num] \rightarrow Num$$

$$f.[] . x1 = \text{minBound}$$

$$f.xs.[] = \text{minBound}$$

$$f.(x \triangleright xs).(y \triangleright ys) = (x - y) \min f.xs.ys$$

21. Derivar una definición recursiva para la función $g: [Char] \rightarrow [Char] \rightarrow Bool$, especificada como sigue:

$$g.xs.ys = \langle \exists as, bs, c, cs : xs = as ++ bs \wedge ys = as ++ (c \triangleright cs) : bs = [] \vee bs.0 < c \rangle$$

Decir en palabras cuándo $g.xs.ys$ es True.

$$xs = c$$

(H)

$$g.xs.ys = \langle \exists as, bs, c, cs : xs = as ++ bs \wedge ys = as ++ (c \triangleright cs) : bs = [] \vee bs.0 < c \rangle$$

(CB)

$$g.[] . ys = \langle \exists as, bs, c, cs : [] = as ++ bs \wedge ys = as ++ (c \triangleright cs) : bs = [] \vee bs.0 < c \rangle$$

$$= \{ \text{Prop concat, eliminación de variable } bs, \text{Def concat} \}$$

$$\langle \exists c, cs : ys = (c \triangleright cs) : [] = [] \vee (bs.0 < c) \rangle$$

$$= \{ \text{lógica, Absorbente } \vee \}$$

$$\boxed{\text{True}}$$

③ $g.xs.C \equiv \langle \exists a_s, b_s, c, cs : \overset{(C \supset C)}{xs = as ++ bs \wedge C = as ++ (C \supset cs) : bs = [] \vee bs.0 < C} \rangle$

= { Prop de concat, eliminación de variable,

$$\langle \exists b_s, c, cs : \underline{xs = bs ++ C} = : (C \supset cs) : bs = [] \vee bs.0 < C \rangle$$

= { lógica, absorbente del $\wedge$, Rango Vacío?

False

④ $g.(x \supset xs)(y \supset ys) \equiv \langle \exists a_s, b_s, c, cs : \underline{(x \supset xs) = as ++ bs \wedge (y \supset ys) = as ++ (C \supset cs) : bs = [] \vee bs.0 < C} \rangle$

= { lógica, Distributiva, Partición de rango }

• $\langle \exists a_s, b_s, c, cs : (x \supset xs) = (as ++ bs \wedge as = []) \wedge (y \supset ys) = (as ++ (C \supset cs) \wedge as = []) : bs = [] \vee bs.0 < C \rangle \vee$ ④

④ $\langle \exists a_s, b_s, c, cs : (x \supset xs) = (as ++ bs \wedge as \neq []) \wedge (y \supset ys) = (as ++ (C \supset cs) \wedge as \neq []) : bs = [] \vee bs.0 < C \rangle$

• $\langle \exists a_s, b_s, c, cs : (x \supset xs) = (as ++ bs \wedge as = []) \wedge (y \supset ys) = (as ++ (C \supset cs) \wedge as = []) : bs = [] \vee bs.0 < C \rangle$

= { eliminación de variable, Def concat }

$\langle \exists b_s, c, cs : \underline{(x \supset xs) = bs \wedge (y \supset ys) = (C \supset cs) : bs = [] \vee bs.0 < C} \rangle \vee$ ④

= { Prop Constructor }

$\langle \exists b_s, c, cs : \underline{(x \supset xs) = bs \wedge x = c \wedge xs = cs : bs = [] \vee bs.0 < C} \rangle$

= { eliminación de variable, lógica, Rango Unitario }

$(x < y) \vee \langle \exists a_s, b_s, c, cs : (x \supset xs) = (as ++ bs \wedge as \neq []) \wedge (y \supset ys) = \overset{a \supset C \supset D (as ++ cs)}{(as ++ (C \supset cs) \wedge as \neq [])} : bs = [] \vee bs.0 < C \rangle$

= { cambio de variable, Def concat, Def constructor }

$(x < y) \vee \langle \exists a_s, b_s, c, cs : \underline{x = c \wedge xs = as ++ bs \wedge y = c \wedge ys = as ++ (C \supset cs) : bs = [] \vee bs.0 < C} \rangle$

= { eliminación de variable }

$(x < y) \vee g.xs.xs$

$g.[Cons] \rightarrow \{Cons\} \rightarrow Bool$

$g.[], xs \equiv true$

$g.xs.C \equiv false$

$g.xs.xs \equiv (x < y) \vee g.xs.ys$

CR) $g \cdot (x \triangleright y) (y \triangleright z) \doteq \langle \exists a, b, c, d : (x \triangleright y) \rangle$

29. Expresar utilizando cuantificadores las siguientes sentencias del lenguaje natural:

- a) El elemento e ocurre un número par de veces en la lista xs .
- b) El elemento e ocurre en las posiciones pares de la lista xs .
- c) El elemento e ocurre únicamente en las posiciones pares de la lista xs .
- d) Si e ocurre en la lista xs , entonces l ocurre en alguna posición anterior en la misma lista.
- e) Existe un elemento de la lista xs que es estrictamente mayor a todos los demás.
- f) Cualquier valor x que anula la función f (es decir que $f.x = 0$) ocurre en la lista xs .
- g) En la lista xs solo ocurren valores que anulan la función f .

a) $\langle \exists i : 0 \leq i < \#xs \wedge xs.i = e : 1 \rangle \bmod 2 = 0$

b) $\langle \forall i : 0 \leq i < \#xs \wedge i \bmod 2 = 0 : xs.i = e \rangle$

c) $\langle \forall i : 0 \leq i < \#xs \wedge i \bmod 2 = 0 \wedge xs.i = e : \langle \forall j : 0 \leq j < \#xs : j \bmod 2 = 1 : xs.j \neq e \rangle \rangle$

d) $\langle \exists i : 0 \leq i < \#xs : xs.i = e \rangle \Rightarrow \langle \forall l : 0 \leq l < \#xs \wedge l < i : xs.l \rangle$

e) $\langle \exists i : 0 \leq i < \#xs \wedge \underline{xs.i} : \langle \forall j : 0 \leq j < \#xs : xs.i < xs.j \rangle \rangle$

f) $\langle \forall i : 0 \leq i < \#xs : xs.i = x \rangle \Rightarrow f(x) = 0$

g) $\langle \forall i : 0 \leq i < \#xs \wedge xs.i : \overbrace{f.x}^{f.x=0} \rangle$

b) $f.xs = \langle Nas, bs, cs : xs = as ++ bs ++ cs : \underline{8 = sum.bs} \rangle$
 (cantidad de segmentos que suman eso).

$$xs = [4, 11, 3, 0]$$

$$f.[4, 11, 3, 0] = \langle \textcircled{Nas}, bs: [4, 11, 3, 0] = \overbrace{as ++ bs}^{xs} : 8 = sum.bs \rangle$$

$$\langle Nas, bs: [4, 11, 3, 0] = \underline{\quad} : 8 = \underline{sum.bs} \rangle$$

= { Aplicacion rango al termino. }

$$8 = \underline{sum.C} + 8 = \underline{sum.4} + 8 = \underline{sum.(11, 1)} + 8 = \underline{sum[4, 11, 3]} + 8 = \underline{sum[4, 11, 3, 0]}$$

{ logica }

$$0 + 0 + \boxed{1} + 0 + 0 = \boxed{1}$$

$$c \in \{0, 1, 2, 3\}$$

$$a = xs.0 \vee a = xs.1 \vee a = xs.2$$

④

bs	as
C	[4, 11, 3, 0] ✓
[4]	[4, 3, 0] ✓
[4, 11]	[3, 0] ✓
[11, 11, 3]	[0] ✓
[4, 11, 3, 0]	C ✓

a) $f: \text{Num} \rightarrow \text{Nat} \rightarrow \text{Num}$ computa la suma de potencias de un número, esto es

$$f.x.n = \langle \sum i : 0 \leq i < n : x^i \rangle$$

b) $\cos: \text{Nat} \rightarrow \text{Num} \rightarrow \text{Num}$ computa la aproximación de la función trigonométrica coseno:

$$\cos.n.x = \langle \sum i : 0 \leq i \leq n : (-x)^{2*i} / (2*i)! \rangle$$

a) $f.x.n = \langle \sum i : 0 \leq i < n : x^i \rangle$

CB $f.x.0 = \langle \sum i : 0 \leq i < 0 : x^i \rangle$

= {lógica, Rango vacío}
[0]

CR $f.x.(n+1) = \langle \sum i : 0 \leq i < n+1 : x^i \rangle$

= {lógica, Partición de rango}

$\langle \sum i : i = n+1 : x^i \rangle + \langle \sum i : 0 \leq i < n : x^i \rangle$

= {HI}

$\langle \sum i : i = n+1 : x^i \rangle + f.x.n$

= {Rango unitario}

$x^{n+1} + f.x.n$

= {Modularización}

$\exp.x.n \equiv x^n$

CB $\exp.n.0 \equiv 1$

CR $\exp.x.(n+1) \equiv x^n \cdot x$

= {HI}

$\exp.x.n \cdot x$

= {Def exp}

$\exp.x.n \cdot x + f.x.n$

=>

$f: \text{Num} \rightarrow \text{Nat} \rightarrow \text{Num}$

$f.x.0 = 0$

$f.x.(n+1) \equiv \exp.x.n \cdot x + f.x.n$

$\exp: \text{Num} \rightarrow \text{Nat} \rightarrow \text{Num}$

$\exp.x.0 \equiv 1$

$\exp.x.(n+1) \equiv \exp.x.n \cdot x$

b) $\cos: \text{Nat} \rightarrow \text{Num} \rightarrow \text{Num}$ computa la aproximación de la función trigonométrica coseno:

$$\cos.n.x = \langle \sum i : 0 \leq i \leq n : (-x)^{2*i} / (2*i)! \rangle$$

$\cos.n.x = \langle \sum i : 0 \leq i < n : (-x)^{2*i} / (2*i)! \rangle$

CB $\cos.0.x = \langle \sum i : 0 \leq i < 0 : (-x)^{2*i} / (2*i)! \rangle$

= {lógica, Rango vacío}

[0]

CR $\cos.(n+1).x = \langle \sum i : 0 \leq i < n+1 : (-x)^{2*i} / (2*i)! \rangle$

= {lógica, Partición de rango}

$\langle \sum i : i = n+1 : (-x)^{2*i} / (2*i)! \rangle + \langle \sum i : 0 \leq i < n : (-x)^{2*i} / (2*i)! \rangle$

= {HI}

$\langle \sum i : i = n+1 : (-x)^{2*i} / (2*i)! \rangle + \cos.n.x$

= {Rango unitario}

$\frac{(-x)^{2n+2}}{(2n+2)!} + \cos.n.x$

= {Def exp, Def factorial}

$\exp.(-x).(2n+2) / f(2n+2) + \cos.n.x$

31. Sea fib la definición recursiva estándar para la función de Fibonacci. Calculá la función de Fibolucci, $Fbl : Nat \rightarrow Nat$, especificada como:

$$Fbl.n = \langle \sum i : 0 \leq i < n : fib.i \times fib.(n-i) \rangle$$

$$Fbl.n \equiv \langle \sum i : 0 \leq i < n : fib.i \cdot fib.(n-i) \rangle \quad (H)$$

$$(CB) Fbl.0 \equiv \langle \sum i : 0 \leq i < 0 : fib.i \cdot fib.(n-i) \rangle$$

= {Rango vacío, lógica}

TD

$$(H) fib.n \equiv fib(n-2) + fib(n-1)$$

$$(CB) fib.0 \equiv 0$$

$$fib.1 \equiv 1$$

$$(CR) fib(n+1) \equiv fib.n + fib(n-1) \quad \checkmark$$

$$(CR) Fbl(n+1) \equiv \langle \sum i : 0 \leq i < n+1 : fib.i \cdot fib.(n-i) \rangle$$

= {lógica, Partición de rango}

$$\langle \sum i : i = n+1 : fib.i \cdot fib.(n-i) \rangle + \langle \sum i : 0 \leq i < n : fib.i \cdot fib.(n-i) \rangle$$

= {HI}

$$\langle \sum i : i = n+1 : fib.i \cdot fib.(n-i) \rangle + Fbl.n$$

= {Rango unitario}

$n - (n+1)$

$$fib.(n+1) \cdot fib.(-1) + Fbl.n$$

$$segf.xs \equiv \langle \exists as, bs : xs = as ++ bs : xs = bs \rangle \quad (H)$$

$$(CR) segf.(x \smallfrown xs) \equiv \langle \exists as, bs : (x \smallfrown xs) = as ++ bs : (x \smallfrown xs) = bs \rangle$$

= {lógica, Distributiva, Partición de rango}

$$\langle \exists as, bs : (x \smallfrown xs) = as ++ bs \wedge as = [] : (x \smallfrown xs) = bs \rangle \vee \langle \exists as, bs : (x \smallfrown xs) = as ++ bs \wedge as \neq [] : (x \smallfrown xs) = bs \rangle$$

= {eliminación de variable as, Def concat}

$$\langle \exists bs : (x \smallfrown xs) = bs : (x \smallfrown xs) = bs \rangle \vee \langle \exists as, bs : (x \smallfrown xs) = as ++ bs \wedge as \neq [] : (x \smallfrown xs) = bs \rangle$$

= {Modularización}

$$msegf.xs \equiv \langle \exists bs : xs = bs : xs = bs \rangle \quad (H)$$

$$(CB) msegf.[] \equiv \langle \exists bs : [] = bs : [] = bs \rangle$$

= {Rango unitario, lógica}

True

$$(CR) msegf.(x \smallfrown xs) \equiv \langle \exists bs : (x \smallfrown xs) = bs : (x \smallfrown xs) = bs \rangle$$

= {lógica, Distributiva, Partición de rango}

$$\langle \exists bs : (x \smallfrown xs) = bs \wedge bs = [] : (x \smallfrown xs) = bs \rangle \vee \langle \exists bs : (x \smallfrown xs) = bs \wedge bs \neq [] : (x \smallfrown xs) = bs \rangle$$

= {eliminación de variable, Rango vacío}

False $\vee$

$$\langle \exists bs : (x \smallfrown xs) = bs \wedge bs \neq [] : (x \smallfrown xs) = bs \rangle$$

= {newto del v}

$$\langle \exists bs: (x \text{D} x_s) = bs \wedge bs \neq [] : (x \text{D} x_s) = bs \rangle$$

= {cambio de variable $bs \leftarrow bs \text{ } bs$, Prop del constructor, eliminación de variable}

$$\langle \exists bs: x_s = bs : (bs \text{D} x_s) = bs \text{ } bs \rangle$$

= {HI}

$$\text{mseg. } x_s \quad \vee \quad \langle \exists as, bs: (x \text{D} x_s) = as ++ bs \wedge as \neq [] : (x \text{D} x_s) = bs \rangle$$

= {cambio de variable, Def concat, Prop del constructor}

$$\text{mseg. } x_s \quad \vee \quad \langle \exists as, bs: x = a \wedge x_s = as ++ bs : (x \text{D} x_s) = bs \rangle$$

= {eliminación de variable}

$$\text{mseg. } x_s \quad \vee \quad \langle \exists as, bs: x_s = as ++ bs : (a \text{D} x_s) = bs \rangle$$